Archives of Computational Methods in Engineering
https://doi.org/10.1007/s11831-025-10372-y

REVIEW

Breast Cancerous Tumor Classification: A Comparative Analysis of Machine Learning and Deep Learning on Different Datasets

Vishnu Vardhana Reddy Karna[1] · Viswavardhan Reddy Karna[2] · Ravinder Beemagani[3] · Varaprasad Janamala[4] · V. N. Koteswara Rao Devana[1] · K. S. Rajasekhar[5] · Ch. V. Ravi Sankar[1] · Padarti Vijaya Kumar[1]

Received: 12 January 2025 / Revised: 25 March 2025 / Accepted: 16 August 2025
© The Author(s) under exclusive licence to International Center for Numerical Methods in Engineering (CIMNE) 2025

Abstract

Breast cancer is a prevalent health issue among women, with one in eight women succumbing to the disease. A significant number of women neglect the necessity for breast cancer detection, as the treatment poses risks associated with exposure to radioactive radiation. Non-invasive procedures, hazardous radiation exposure, and limited diagnostic specificity for breast tumors hinder breast cancer screening methods. Currently, several medical technologies, including mammography, magnetic resonance imaging, computed tomography, positron emission tomography, and histopathological imaging, are being used to diagnose breast cancer early. Nevertheless, proficient radiologists or pathologists are necessary to interpret these imaging modalities. The process is challenging, costly, and prone to inaccuracies. Advances in machine learning and computing technology have transformed many facets of the world in the last ten years. Deep learning models have emerged as powerful tools in detecting tumors and predicting breast cancer, leveraging radiographic and histopathological images to achieve remarkable outcomes. Nevertheless, rigorous external validation is essential to ensure these advanced artificial intelligence technologies can be confidently integrated into clinical decision-making processes. The main objective of this research is to provide a critical analysis of findings on breast cancerous tumor classification using machine and deep learning algorithms. For this, we examined five prominent datasets: Wisconsin, SEER, ultrasound pictures, mammograms, and BreakHis histopathology images. The study reviewed the literature from the past decade (2015–2024) across multiple sources, including Springer, ScienceDirect, IEEE, PubMed, MDPI, Nature, Web of Science, Hindawi, and ArXiv. A detailed discussion was incorporated to elucidate ongoing research challenges and opportunities for future research in this burgeoning field.

✉ Vishnu Vardhana Reddy Karna
vishnuvardhanareddyk@adityauniversity.in

Viswavardhan Reddy Karna
viswavardhank@rvce.edu.in

Ravinder Beemagani
ee21d003@iittp.ac.in

Varaprasad Janamala
varaprasad.janamala@christuniversity.in

V. N. Koteswara Rao Devana
koteswararaodevana@adityauniversity.in

K. S. Rajasekhar
rajsekharkotra@gmail.com

Ch. V. Ravi Sankar
venkataravisankarch@adityauniversity.in

Padarti Vijaya Kumar
vijayakumarp@adityauniversity.in

1 Department of Electronics and Communication Engineering, Aditya University, Surampalem 533437, India

2 Department of Artificial Intelligence and Machine Learning, R V College of Engineering, Bengaluru 560059, India

3 Department of Electrical Engineering, Indian Institute of Technology, Tirupati 517619, India

4 Department of Electrical and Electronics Engineering, Christ University, Bengaluru 560074, India

5 Department of Electronics and Communication Engineering, Acharya Nagarjuna University, Guntur 522510, India

Published online: 05 September 2025

1 Introduction

Breast cancer is a condition characterized by the uncontrolled proliferation of aberrant breast cells, resulting in tumor formation. The tumors have the potential to grow all over the human body and become lethal if untreated. Breast cancer cells start inside the breast's milk ducts and/or milk-producing lobules. The initial form (in situ) is non-life-threatening and can be identified in its initial stages. Cancer cells can infiltrate adjacent breast tissue. This induces tumors that result in thickening or lumps. Invasive tumors can metastasize to adjacent lymph glands or other body parts. Metastasis can be perilous and fatal. The patient, the cancer kind, and the extent of the disease determine treatment. The treatment encompasses surgical intervention, chemotherapy, and pharmacological agents. Non-invasive breast tumors do not spread to neighboring lobules or ducts; instead, the cancer cells remain contained within the impacted breast area. In the instance of breast carcinoma, the malignancy has not metastasized beyond the localized region. There are two forms of in situ cancer: lobular carcinoma in situ (LCIS) and ductal carcinoma in situ (DCIS). DCIS arises when a neoplasm develops within the duct that transports milk from the lobules to the nipple. Nonetheless, LCIS may stimulate the proliferation of cells that mimic malignant ones. Hence, it is crucial to manage it as though it were cancerous. The majority of LCIS instances stay localized. However, doctors may choose to continue monitoring their patients as a precaution against invasive breast cancer. For breast cancer to be successfully treated, early cancer identification is essential.

Consequently, the accessibility of suitable screening technologies is not just important but essential for detecting the initial indications of breast cancer. Early identification of cancer can be life-saving, making the urgency of this issue clear [1, 2].

Machine learning (ML) is a powerful tool in the fight against cancer. It is a subset of artificial intelligence that utilizes diverse statistical, probabilistic, and optimization methodologies. It enables computers to gain insight from historical data and identify subtle patterns within extensive, noisy, or intricate datasets. Multiple research studies have employed machine learning to forecast the prognosis of various cancer patients, including those with breast cancer, lung cancer, and liver cancer. The machine learning model has consistently exhibited a commendable, individualized predictive ability [3, 4]. In the past decade, computer technology and machine learning advancements have revolutionized various aspects of the world. Deep learning (DL), a subset of ML, has demonstrated exceptional performance across multiple domains, especially in the biomedical sector, due to its capacity to process extensive datasets [5, 6]. While the capabilities of deep learning models have been employed and assessed in the detection and prognosis of breast cancer, employing radiographs and histopathology images, and have demonstrated commendable performance, it's important to note that AI has demonstrated favorable assertions solely in retrospective research. Implementing these advanced AI technologies as clinical decision-making instruments requires rigorous external validation [7, 8].

1.1 Current Statistics of Breast Cancer

Breast cancer is not just a local or national issue but a global one. It is one of the most often reported cancers and is recognized as a leading cause of mortality among females worldwide. When it comes to breast cancer, being a woman is the most important determinant [9, 10]. Men account for just 0.5–1% of all breast cancer cases, whereas women account for over 99%. In 2022, 2.3 million women received a breast cancer diagnosis, and 670,000 people died from the disease worldwide. The disease affects women globally, occurring at any age post-puberty, with incidence rates rising in older age groups. The burden of breast cancer varies significantly depending on human growth and development, based on estimates from around the world. For example, 1 in 12 women may receive a breast cancer diagnosis over their lifetime, and 1 in 71 will pass away from the disease in nations with highly high Human Development Indexes [11].

Breast carcinoma is the most prevalent cancer identified in women in the United States. Annually, about one-third of all newly identified malignancies in women are attributed to breast cancer. In 2024, an estimated 310,720 women will receive a diagnosis of invasive breast cancer, with 16% of cases occurring in women under the age of 50. Additionally, 56,500 women will receive a diagnosis of DCIS, or ductal carcinoma in situ. Even though it is uncommon, males can develop breast cancer; their lifetime risk is less than 1%. According to studies, there will be roughly 530 male breast cancer deaths and 2,790 new instances of progressive breast cancer diagnosed in men in 2024 [12–14]. Two Female hormone-responsive tumors that are strongly related to one another are ovary and breast cancers. Around 5 to 10% of newly discovered breast cancer women possess a familial history of breast or ovary cancer, indicating the influence of either genetic or non-genetic heredity on the formation of breast cancer [15, 16]. Figure 1 illustrates the distribution of the top ten predominant cancers in Indian women – 2022.

From Fig. 1, the top five predominant cancers among Indian women were breast with 28.8%, cervix with 10.6%, ovary with 6.2%, corpus uteri with 3.7%, and lung with 3.7%. In India, there will probably be 1.57 million cancer cases in 2025 compared to 1.46 million in 2022 [17].

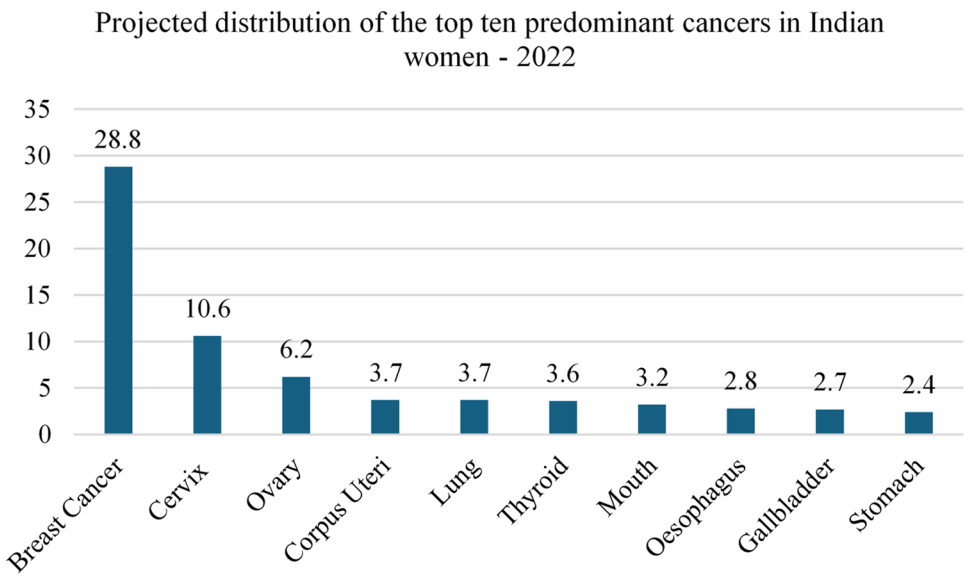


Fig. 1 Projected distribution of the top 10 predominant cancers in Indian women − 2022

1.2 Types of Breast Cancer

Invasive breast cancers (IBC) encompass a diverse range of tumors that exhibit variability in their medical manifestations, behavior, and appearance. Previously, IBCs, referred to as invasive ductal carcinoma, constitute the most prevalent subgroup, accounting for 40–80% of cases. This type is inherently diagnosed as a tumor that cannot be categorized into any specific histological subtype. The majority consensus clustering validated the identification of four primary breast cancer inherent subtypes based solely on messenger Ribonucleic Acid (mRNA) gene expression levels: Luminal A, Luminal B, human epidermal growth 2 (HER2), and triple-negative breast cancer (TNBC) [18, 19].

Luminal A tumors are defined by the presence of estrogen receptors (ER) and/or progesterone receptors (PR), less than 20% Ki67, and lack of human epidermal growth factor receptor 2 (HER2). Clinically, they have the most significant prognosis, with a greater survival rate and a lower relapse rate. They are also low-grade and slow-developing. These cancers respond well to hormone therapy, but chemotherapy has less effect [20].

Luminal B tumors exhibit an improved severity and a poorer prognosis than Luminal A tumors. They are estrogen receptor-positive, maybe progesterone receptor-negative, and exhibit more than 20% of Ki67. These tumors may respond favorably to hormone therapy in conjunction with chemotherapy [21].

The HER2-positive cohort comprises 10–15% of breast tumors, distinguished by elevated levels of HER2 and a lack of estrogen receptor and progesterone receptors. They exhibit accelerated growth compared to luminal variants, and the prognosis has enhanced following the implementation of HER2-targeted therapy. The HER2-positive variant is characterized by increased aggressiveness and rapid proliferation [22].

The absence of estrogen, progesterone, and human epidermal growth factor receptors characterizes triple-negative BC. They represent approximately 20% of all breast carcinoma cases. It is predominantly observed in women under the age of 40 years, particularly among African American women [23]. Table 1 summarizes the breast cancer types, characteristics, symptoms, diagnostic tests, and treatment.

Table 1 categorizes breast cancer into four major types based on receptor status (ER, PR, HER-2) and other molecular markers. Each type has specific clinical symptoms, diagnostic approaches, and tailored treatments. For instance:

- **Luminal types (A and B)**: Hormone-receptor positive, with hormone therapy and chemotherapy treatments.
- **HER-2 positive**: Treated with HER-2 targeted therapies.
- **Triple-Negative**: Aggressive, often treated with a combination of therapies due to the lack of hormone receptor targets.

1.3 Risk Factors and Symptoms of Breast Cancer

Breast cancer is more likely in women who are overweight, have a personal or family history of the disease, have been exposed to radiation in the past, have an entire reproductive history (including the onset of menstruation and the age of their first pregnancy), tobacco usage, excessive alcohol consumption, and use of postmenopausal hormones. Aside from being female and being over the age of 40, there is no known cancer-causing risk factor for almost half of the breast cancers that form in women [24]. Breast cancer is more common in families where the disease has a history. However, the majority of women who get the disease do

Table 1 Breast cancer types, characteristics, symptoms, diagnostic test, and treatment

Breast cancer type	Characteristics	Symptoms	Diagnostic test	Treatment
Luminal A	Positive for ER, Positive for PR, Negative for HER-2, Low Ki-67	Breast pain, nipple discharge, thickening, swelling	Mammogram, MRI, Ultrasound	Radiation therapy, Chemotherapy, Surgery
Luminal B	Positive for ER, Negative for PR, Positive for HER-2, High Ki-67	Rubbery lump, pain in one or both breasts, skin dimpling, nipple discharge	Mammogram, MRI, Ultrasound, Breast biopsy	Radiation therapy, Targeted therapy, Chemotherapy, Surgery
HER-2	HER-2 gene mutation	Change in size and shape of breast, lump or mass, fluid discharge,	Immunohistochemistry, Fluorescence in situ hybridization	Targeted therapy, Chemotherapy
Triple Negative BC	Absence of ER, PR, HER-2	Nipple retraction, swelling breast, dimpling breast skin, inflamed lymph nodes	Mammogram, MRI, Ultrasound, CT scan, PET scan	Chemotherapy, Immune therapy, Surgery, Targeted and radiation therapy

not come from a family where it has been diagnosed [25]. Around 15% of women diagnosed with breast cancer have a familial history of the disease. It is not always the case that a woman's risk is lower just because there is no known family history. Approximately 5–10% of breast cancer cases are related to identifiable genetic modifications transmitted from either parent. Most notably, mutations in the BRCA1, BRCA2, and PALB-2 genes substantially raise the risk of breast cancer, which is a hereditary disease with a high penetrance. Women who are identified with mutations in genes may think about risk-reduction measures like chemoprevention or surgically removing both breasts. Most individuals will not exhibit any apparent signs during the early stages of cancer, which emphasizes the significance of early detection [26, 27]. Figure 2 displays the risk factors of breast cancer disease.

Breast cancer may present a variety of symptoms, particularly in its advanced stages. Possible manifestations of breast cancer may encompass a breast mass or induration, frequently asymptomatic alteration in size, contour, or morphology of the breast dimpling, erythema, pitting, or other dermal modifications modification in nipple morphology or the peri nipple skin atypical or sanguineous discharge from the nipple. Individuals with an atypical breast lump should get medical attention, regardless of the absence of pain. Many breast lumps are non-cancerous. There is a greater chance of successful treatment for malignant breast tumors that are tiny and have not migrated to neighboring lymph nodes. Breast cancer can cause additional symptoms and spread to other parts of the body. The most frequently observed initial location of metastasis is the lymph nodes

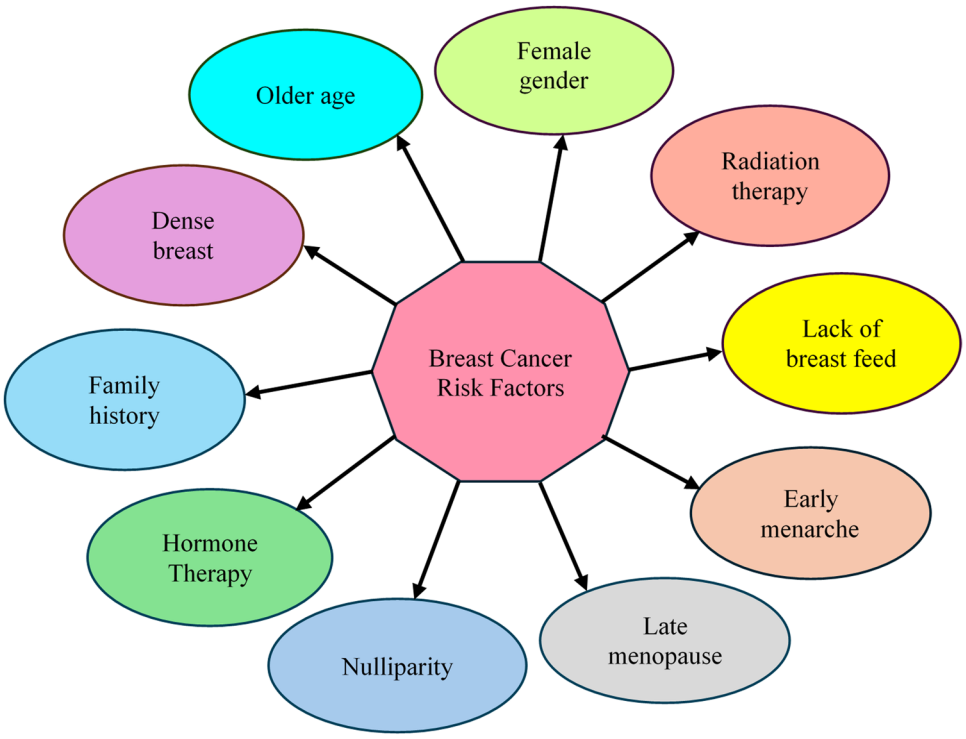


Fig. 2 Risk factors of breast cancer disease

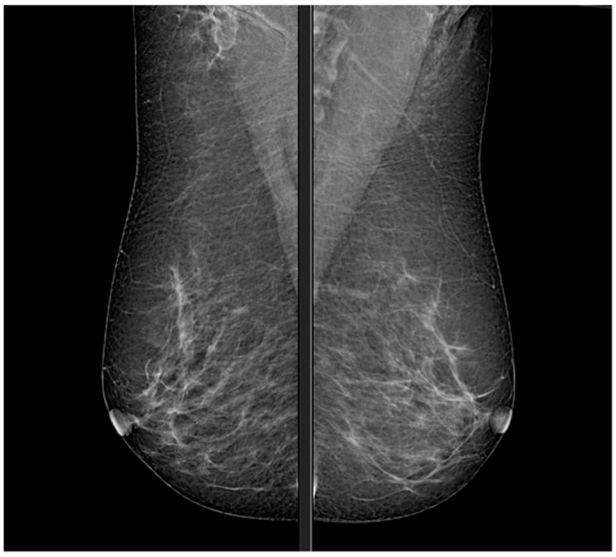


Fig. 3 Breast Mammogram [34]

in the axillary region; however, it may be feasible to have cancerous lymph nodes that are not palpable [28, 29].

1.4 Conventional Brest Cancer Diagnostic Tests

Patients with breast tumors typically have a variety of tests performed on them, such as mammography, magnetic resonance imaging (MRI), computed tomography (CT), and ultrasonography. After every examination, the patient's disease condition will be described in an informal clinical report. Clinicians typically make diagnostic decisions for patients with highly suspected malignant breast cancer after thoroughly considering various medical data [30–32].

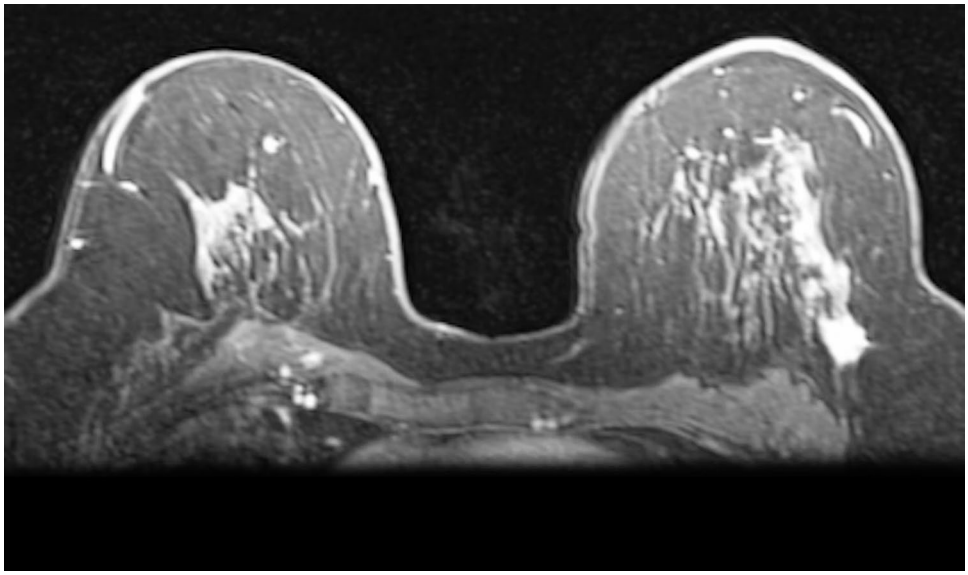
1.4.1 Mammogram

Mammography is usually considered the most effective method for breast cancer diagnosis [4]. Each breast has a low-dose X-ray examination, which looks for breast tumors, skin abnormalities, and thickening or discharge of the nipples. A mammogram is a digital image taken vertically and horizontally to cover a breast fully. These mammograms assess for masses/lesions, calcified tissue, and structural aberrations in the breast area. Mammography identifies masses or lesions at a point when they have already developed into tumors. The findings are documented in a Breast Imaging Reporting and Data System (BI-RADS) to ease communication with the clinician for additional evaluations [33]. Figure 3 shows the breast mammogram.

1.4.2 Magnetic Resonance Imaging (MRI)

MRI is a conventional method that utilizes a magnetic field, unlike X-rays in a mammogram, to examine breast masses that subsequently require biopsy. MRI is a supplementary tool to mammography and ultrasound in breast cancer screening, enhancing detection rates. Technological advancements in MRI techniques facilitate the assessment of metabolic variability within tumors. Nevertheless, the necessity of advanced equipment limits accessibility to all hospitals and clinics. In contemporary clinical practice, MRI is not utilized for screening purposes but mainly to delineate tumor size and/or identify other tumor sites [35, 36]. Figure 4 shows the breast MRI image.

Fig. 4 Breast MRI image [37]



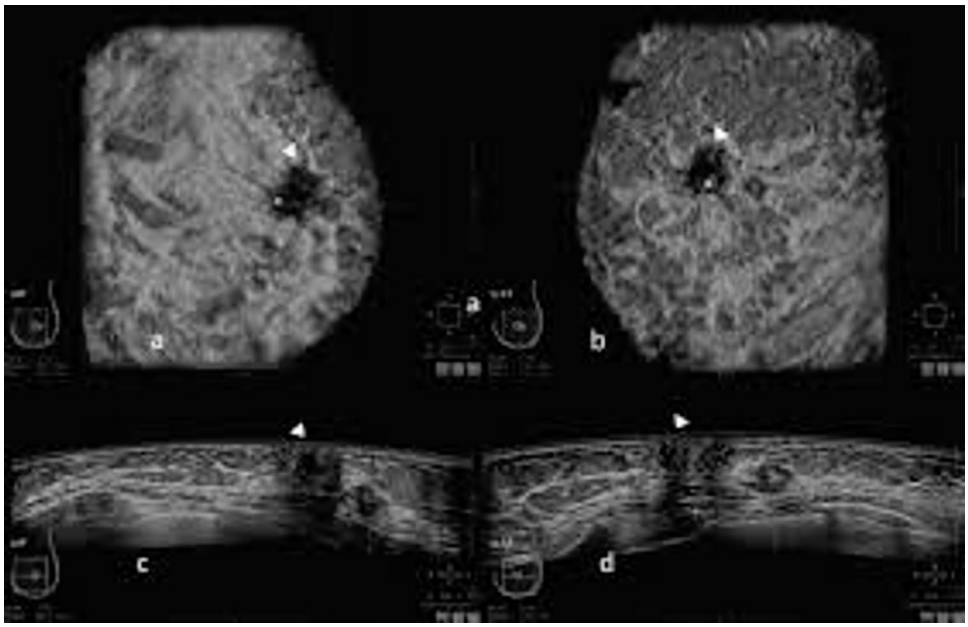


Fig. 5 Breast ultrasound image in 3D [40]

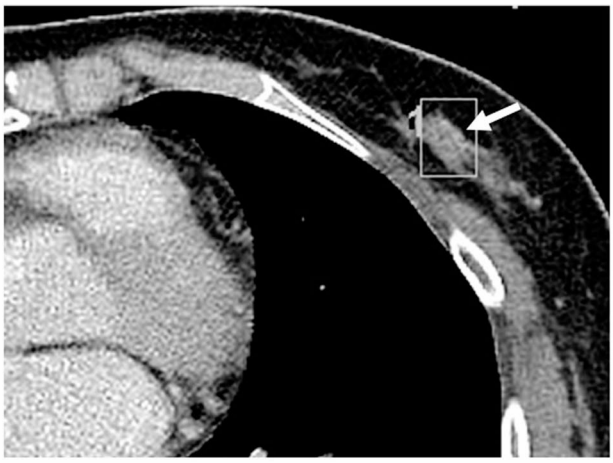


Fig. 6 Breast CT scan image [43]

1.4.3 Ultrasound Imaging

Ultrasound serves as a supplementary treatment to validate a positive mammography. In a breast ultrasound, sound waves are used to produce images of the breast's inside. It is frequently employed to ascertain if a lump in the breast is a solid tumor or a cyst, i.e., filled with fluid, and it may also help determine the direction of a biopsy probe. Because ultrasound imaging has higher sensitivity than a mammogram, the research proves that it is a more efficient strategy for young women with thick breast tissue. However, because of its poor specificity, more biopsies for benign tumors might be performed [38, 39]. Figure 5 shows the breast ultrasound image in 3D.

1.4.4 Positron Emission Tomography/Computed Tomography

Positron Emission Tomography (PET) delivers insights into tumor anatomy at a molecular scale using radiotracers, whereas Computed Tomography (CT) generates images containing anatomical information. PET/CT denotes the method that integrates PET and X-ray CT scanners to provide images that pinpoint the anatomical site of abnormal metabolic processes within the body. The PET/CT approach's primary advantage is its integration with both functional and structural imaging, facilitating enhanced visualization of lesions compared to traditional screening methods [41, 42]. Figure 6 shows the breast CT scan image.

1.4.5 Digital Breast Tomosynthesis

Digital Breast Tomosynthesis (DBT) is a sophisticated mammography method that uses low-dose X-ray technology to create a three-dimensional high-resolution reconstruction of breast tissue utilizing numerous pictures captured from thin slices (1 mm) [44]. This technique is frequently employed as a supplementary method to mammography for breast cancer diagnosis following its approval by the US Food and Drug Administration in 2011.

Nonetheless, the primary drawbacks of this technology include the substantial expense of the test, the discomfort associated with the operation, and the utilization of radiation. Integrating 2D and 3D mammography enhances screening efficacy for more invasive tumors [45, 46]. Figure 7 displays the digital breast tomosynthesis image.

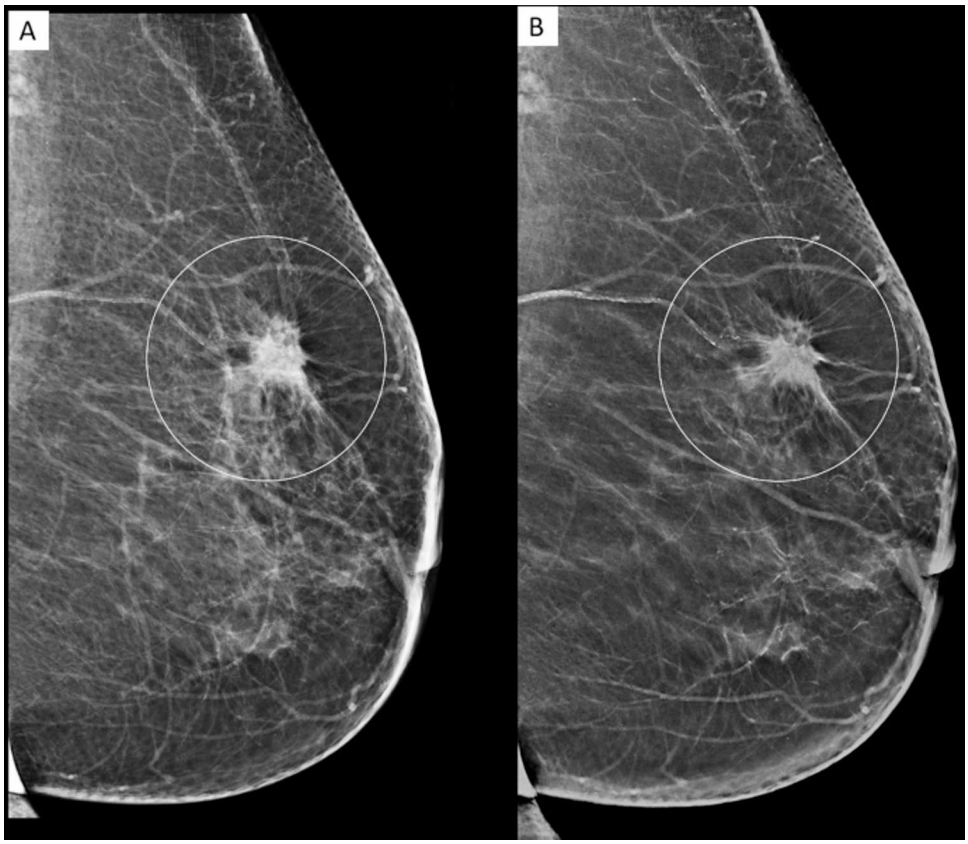


Fig. 7 Digital breast tomosynthesis image [47]

Most commonly used techniques such as mammograms, magnetic resonance imaging (MRI), and ultrasound mainly relied on finding the physical tumor present in the breast. Even though mammography is the gold standard method, several disadvantages still affect the accuracy. Mammography can come with some risks. However, the recommendations on when and how often a person should undergo screening consider these risks. Regular testing means repeated exposure to low levels of radiation, which may increase the risk of cancer slightly. Mammography can also reveal noncancerous changes, leading to false-positive results. This can lead to short-term anxiety, further tests, and treatment, which may be unnecessary and costly. In addition, the results may be less accurate for women with dense breast tissue or breast implants. This is because dense breast tissue can appear white on the mammography image, which can mask the presence of tumors, as these also appear white, resulting in false negatives. Mammography can be uncomfortable since it involves compression of breasts. Also, it causes rupturing of tumor encapsulation due to pressure.

Although MRI can find some cancers not seen on a mammogram, it's also more likely to find things that turn out not to be cancer (called a *false positive*). This can result in a woman undergoing tests and/or biopsies, which are not required. That is why MRI is not recommended as a screening test for women at average risk of breast cancer. Ultrasound is an effective method for identifying cysts and solid tumors. It is used as complementary to mammography since it can be able to examine the dense breasts and increase sensitivity. A new diagnostic tool needs to be designed and improved, focusing on early detection of small tumors for better diagnosis.

The detection and surveillance of breast cancer are the primary components of its treatment. Data from patients and primary tumor characteristics, including tumor size, lymphatic status, tumor grade, and treatment methods, have been utilized to develop prognostic models. Nonetheless, despite significant endeavors in the early identification of recurrent illnesses, research indicates that only a limited fraction of recur cases are recognized during the asymptomatic phase. Multidisciplinary investigation or machine learning is essential for aiding physicians in predicting breast cancer resurgence [48, 49].

With the recent rapid advancements in AI and its use in clinical cancer research, cancer prediction performance has increased to unprecedented levels [50]. Advanced AI methodologies, particularly machine learning, and deep learning may produce medical insights from extensive datasets to facilitate informed medical decisions. These AI methodologies are non-invasive approaches for diagnosing the illness without causing harm to an individual. Machine learning is

regarded as a reliable and accurate approach for synthesizing different data types to enhance diagnostic precision [51, 52].

2 Article Selection

To investigate the literature on the classification of breast cancerous tumors, databases including Springer, Nature, ScienceDirect, IEEEXplore, Web of Science, PubMed, MDPI, and Wiley were queried utilizing keywords such as breast cancer, breast carcinoma, breast cancer prediction, breast cancer tumor classification, benign tumor, malignant tumor, data preprocessing, machine learning algorithms, ensemble classifiers, hybrid algorithms, deep learning algorithms, feature extraction, feature selection, and hyperparameter optimization techniques. This review comprises peer-reviewed academic publications, international conference proceedings, and reviews published in English from 2015 to 2024, incorporating both qualitative and quantitative data. Articles chosen from reputed international journals for this comprehensive review on breast cancerous tumor classification are provided in Table 2.

This study examined literature from 2015 to 2024 about breast cancer classification. A majority of work during the past six years constituted approximately 81%. The pie chart in Fig. 8 shows the year-wise percentage of publications considered in this work.

3 State-Of-The-Art Works

3.1 Using the Wisconsin Breast Cancer Dataset

In biomedical research, the Wisconsin Breast Cancer Dataset is one of the most well-known datasets for breast cancer disease diagnosis using artificial intelligence. It includes details regarding breast cancer tumors, such as characteristics calculated from digital images of a breast mass taken by fine needle aspiration (FNA) method. This dataset includes 569 samples, of which 357 are benign, and 212 are malignant. The multivariate dataset has 30 real-valued input features and one output feature (diagnosis) indicating whether the tumor is benign or malignant. The features include radius, texture, perimeter area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension, among other attributes of the cell nucleus [53].

Naik et al. [54], proposed Binary Bat Algorithm (BBA), a wrapper-based feature selection method, and evaluated the performance of a One-pass Generalized Classifier Neural Network (OGCNN) on selected features from the six UCI machine learning datasets, including the WDBC dataset. It has been concluded that the proposed method outperformed other methods such as Extreme Learning Machine (ELM), Probabilistic Neural Network (PNN), and Radial Basis Function Neural Network (RBFNN). The best accuracy achieved on the WDBC dataset is 93.53%. The authors did not disclose the number of features selected from the dataset using the proposed wrapper approach. The accuracy of the proposed method can be improved further.

Lahoura et al. [55] Proposed a framework for breast cancer disease classification based on cloud computing utilizing extreme learning machine (ELM), a variant of ANN. The proposed framework has been evaluated on both standalone and cloud environments. The research work considers the WBCD dataset and normalized it. Then, the dataset was divided into training sets (80%) and testing sets (20%). The gain ratio feature selection method has been employed to select 14 significant features out of 32 and evaluate different ML models. Achieved an accuracy of 96.92% in a standalone environment, whereas 98.68% in a cloud environment. Accuracy can be further improved. Hyperparameter tuning of ML models has not been employed in this work.

Alshayeji et al. [56] Classified benign and malignant breast cancerous patients using a shallow artificial neural network consisting of a single hidden layer, with 5-fold cross-validation. The authors considered two breast cancer datasets, WDBC (569 instances, 30 features) and WBCD (699 cases, nine features), and splatted them into an 80:20 ratio for training and testing. The model parameters, such as the number of neurons and activation functions, were trained before training and validation—the utmost accuracy of 99.85% on the WBCD dataset and 99.47% on the WDBC dataset. Data pre-processing steps and feature selection techniques were not addressed in this research.

Kadhim et al. [57], evaluated the performance of ten machine learning classifiers (DT, LDA, RF, GB, LR, NB, SVM, passive-aggressive, nearest centroid, and perceptron) on the Wisconsin dataset for early breast cancer diagnosis. The Gradient Boosting algorithm has achieved an accuracy of 97.36% and a specificity and precision of 100%. Data pre-processing steps were not discussed, and attribute selection techniques were not employed in this work.

Singh et al. [58] Applied various strategies for the early diagnosis of breast cancer using machine learning. Classification of breast cancer tumors was conducted using C 5.0, Naive Bayes, logistic regression, random forest, ctree, KNN, K-Means, GBM, AdaBoost, and decision tree models, with performance evaluations on the Wisconsin and SEER datasets. The classifiers' performances were evaluated using accuracy, precision, recall, and F1 scores. Additionally, it forecasted the tumor's viability, considering tumor size, different cancer phases, and months of living. In the Wisconsin dataset, a classification accuracy of 97.66%

Table 2 Articles chosen from reputed international journals

S.No.	Source	Journal name
1.	Springer	Archives of Computational Methods in Engineering
		Interdisciplinary Sciences: Computational Life Sciences
		Journal of Ambient Intelligence and Humanized Computing
		Multimedia Tools and Applications
		Health Information Science and Systems
2.	Nature	Scientific Reports
3.	ScienceDirect	Measurement
		Expert Systems with Applications
		Journal of Radiation Research and Applied Sciences
		Biomedical Signal Processing and Control
		Journal of Computational Science
		Heliyon
		Computerized Medical Imaging and Graphics
		Medical Image Analysis
		Computer Methods and Programs in Biomedicine
		Engineering Applications of Artificial Intelligence
		Ultrasound in Medicine & Biology
		The Breast
		Radiography
		Journal of Biomedical Informatics
		Computers in Biology and Medicine
		Progress in Biophysics and Molecular Biology
		Intelligence-Based Medicine
		Critical Reviews in Oncology/Hematology
		Materials Today: Proceedings
4.	IEEEXplore	IEEE Transactions on Medical Imaging
		IEEE Access
		IEEE International Conferences
5.	MDPI	Applied Sciences
		Diagnostics
		Cancers
		Biology
		Health Care
6.	Wiley	Journal of Sensors
		BioMed Research International
		International Journal of Imaging Systems and Technology
7.	Others	ACM Transactions on Internet Technology
		PLOS ONE
		Journal of Biomedical Physics and Engineering
		Journal of Translational Medicine
		Frontiers in Pharmacology
		Frontiers in Public Health
		Computational Intelligence, Machine Learning and Data Analytics
		cancer Medical Science
		Breast Cancer: Targets and Therapy

was achieved with RF and SVM classification algorithms, while the SEER dataset got a maximum accuracy of 99.5% utilizing a Decision Tree. The data pre-processing steps and feature selection method employed in this work are not discussed adequately.

Elsadig et al. [59] (2023) utilized five feature selection strategies (Chi-square (X2), ReliefF, Analysis of Variance (ANOVA), Gini Index, and Gain Ratio) and eight classification algorithms (KNN, DT, SVM, RF, MLP, NB, LR, and Stack) to improve the breast cancer detection process. A dataset of 17 salient features has been derived from the WDBC dataset to minimize computational expenses, and the efficacy of machine learning classifiers has been assessed by partitioning the dataset into training and testing sets in ratios of 60:40, 70:30, 80:20, and 90:10, respectively. SVM was found to perform better than other classifiers,

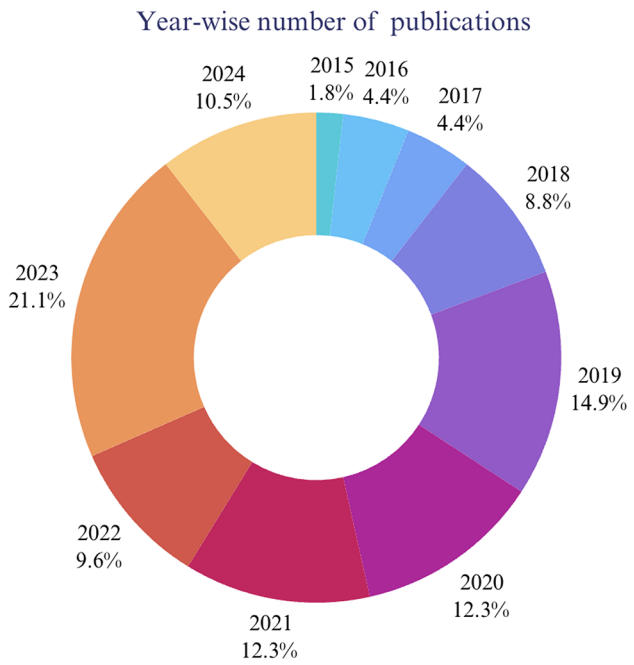


Fig. 8 Year-wise percentage of publications

having an accuracy of 97.7% when employing an 80:20 split on specific features. No pre-processing steps were discussed in the proposed work; accuracy can still be improved.

Reshan et al. [60] Introduced an advanced ensemble machine learning algorithm that integrates voting, stacking, bagging, and boosting techniques for detecting breast cancer. The WDBC dataset has been obtained, and the recursive feature reduction method has been utilized to identify the most informative features. Twenty-two (22) attributes have been selected, and the performance of the proposed ensemble algorithm has been assessed using six performance indicators. It was determined that the stacking strategy emerged as the superior model, with an accuracy of 99.89%. The dataset is imbalanced concerning the target class (63% benign, 37% malignant) and has not been addressed in the research work—lack of optimization and deep learning methods.

Mushtaq et al. [61] suggested integrating the Gaussian Naive Bayes algorithm with a Chi-squared attribute selection technique. This experiment was conducted following the dimensionality reduction of the WBC data using extended Kernel Principal Component Analysis (K-PCA). Five kernels in Kernel Principal Component Analysis (K-PCA) have been evaluated following the execution of requisite pre-processing techniques. Six out of eleven traits have been identified as noteworthy. The suggested system's performance has been assessed using accuracy, precision, sensitivity, and specificity derived from a confusion matrix. The proposed methodology, utilizing six selected features and sigmoid K-PCA, achieved an optimal accuracy of 99.28%. Only one ML algorithm (NB) has been employed in this work. Dataset imbalance issues were not considered.

Hardeep Kaur [62] Introduced a unique dense Convolutional Neural Network (DCNN) deep learning model to reliably classify benign and malignant tumor types utilizing the Wisconsin Breast Cancer dataset. The performance metrics—accuracy, sensitivity, precision, F1-score, and area under the ROC curve (AUC)—were calculated using five-fold cross-validation. The suggested CNN model has been found to detect anomalies related to breast cancer with a good accuracy of 98.2%. The data pre-processing steps adopted were not discussed in detail. Feature selection techniques are also not employed in this research.

Singh et al. [63] Employed three feature selection strategies: Gravitational Search Optimization Algorithm (GSA), Emperor Penguin Optimization (EPO), and a hybrid approach combining GSA and EPO (hGSAEPO) on the WDBC dataset to identify significant features by eliminating irrelevant ones. Six machine learning classifiers—Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and an ensemble of five classifiers—have been utilized on selected variables and their performance assessed using various metrics. A classification accuracy of 98.31% has been attained utilizing a random forest classifier with features picked through a hybrid method. Accuracy can be improved further; no pre-processing steps are discussed.

Awan et al. [64] Conducted a comparative analysis of seven machine learning models, including logistic regression (LR), support vector machine (SVM), k-nearest neighbors classifier (KNN), decision tree classifier (DT), random forest classifier (RF), Naïve Bayes (NB), and artificial neural network (ANN), to predict breast cancer utilizing the Wisconsin breast cancer and breast cancer datasets. During data pre-processing, the median value has been substituted to mitigate the impact of outliers. Using the Wisconsin breast cancer dataset, KNN demonstrated an accuracy of 99%, whereas RF achieved 98%. On the other hand, using the breast cancer (BC) dataset, the highest accuracy was attained by logistic regression (LR) at 83%. In comparison, the lowest was recorded by decision tree (DT) at 65%, indicating that the numeric dataset WBC exhibits superior accuracy compared to the breast cancer dataset. The authors failed to address the issue of dataset class imbalance.

Jony et al. [65] Performed a comparative analysis of the Wisconsin Breast Cancer Diagnostic (WBCD) dataset utilizing four different deep learning models: Feedforward Neural Network (FNN), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU). The efficacy of the DL algorithms has been assessed using accuracy, precision, recall, and F1-score. CNN attained the highest accuracy at 98.25%, succeeded

by GRU at 97.37%, FNN at 96.49%, and LSTM at 95.61%. No pre-processing steps were discussed in the work. Feature selection methods were not employed.

The summary of related works on the Wisconsin Breast Cancer Dataset has been provided in Table 3, which includes the Author name and year of the publication, feature selection technique chosen, no. of features selected by the work, the ML/DL algorithm employed, best accuracy achieved by them, and the limitations of the work.

3.2 Using SEER Breast Cancer Dataset

The Surveillance, Epidemiology, and End Results (SEER) dataset comprises breast cancer medical records from the United States. The SEER registries gather information regarding patient demographics, primary tumor location, tumor shape, diagnosis stage, and the initial therapy course while conducting follow-ups for vital status. The SEER database, maintained by the National Cancer Institute, is the

Table 3 Summary of related works on the Wisconsin breast cancer dataset

Author & Year	Feature Selection technique	No. of features	ML/DL Algorithm	Max. accuracy achieved (%)	Limitations
Naik et al. [55]	Binary Bat Algorithm (BBA)	30	OGCNN	93.53	The authors have not disclosed the number of features selected from the dataset using the proposed wrapper approach.
Lahoura et al. [56]	Gain Ratio	14	Extreme learning machine	98.68	Accuracy can be further improved. Hyperparameter tuning of ML models has not been employed in this work.
Alshayeji et al. [57]	–	30	Shallow ANN	99.85	Data pre-processing steps and feature selection techniques were not addressed in this research.
Kadhim et al. [58]	–	31	Gradient Boosting	97.36	Data pre-processing steps were not discussed, and attribute selection techniques were not employed in this work.
Singh et al. [59]	–	31	RF, SVM	97.66	The data pre-processing steps and feature selection method employed in this work are not discussed adequately.
Elsadig et al. [60]	Chi-square, ReliefF, ANOVA, Gini Index, and Gain Ratio	17	SVM	97.70	No pre-processing steps were discussed in the proposed work; accuracy can still be improved.
Reshan et al. [61]	Recursive Feature Elimination	22	Stacking	99.89	The dataset is imbalanced concerning the target class (63% benign, 37% malignant) and has not been addressed in the research.
Mushtaq et al. [62]	Chi-square Kernel PCA	6	Gaussian Naïve Bayes	99.28	Only one ML algorithm (NB) has been employed in this work. Dataset imbalance issues were not considered.
Hardeep Kaur [63]	–	31	Dense CNN	98.20	The data pre-processing steps adopted were not discussed in detail. Feature selection techniques are also not employed in this research.
Singh et al. [64]	Gravitational Search Optimization Algorithm (GSA), Emperor Penguin Optimization (EPO), Hybrid GSAEPO	-	Random Forest	98.31	Accuracy can be improved further; no pre-processing steps are discussed.
Awan et al. [65]	–	31	KNN	99.00	The authors failed to address the issue of dataset class imbalance.
Jony et al. [66]	–	31	Convolutional Neural Networks	98.25	No pre-processing steps were discussed in the work. Feature selection methods were not employed.

preeminent source of statistics regarding cancer incidence and survival in the United States. About 34.6% of Americans are currently covered by population-based cancer registries from which SEER gathers and disseminates data on cancer incidence and survival. The dataset comprises 590,646 samples and 133 features, including race, sex, age at diagnosis, marital status, breast cancer subtype, grade, N stage, M stage, radiation, chemotherapy, surgery, survival months, sequence number, primary site, vital status, and so on [66]. However, only 63,145 patient records were chosen for research purposes, removing those with incomplete data and those whose survival duration was under sixty months while still alive [67]. This dataset was collected between 1973 and 2014 and is available at https://seer.cancer.gov.

Shukla et al. [68] Utilized unsupervised machine learning techniques such as self-organizing map (SOM) and density-based spatial clustering of applications with noise (DBSCAN) to establish nine cohort clusters of individuals based on similar features. These cohorts have been utilized to improve the classification accuracy of breast cancer patients using multilayer perceptron (MLP) with a 10-fold CV. The number of samples employed is 85,189, and the selected attributes are 29. Achieved training accuracy of 93.53% and testing accuracy of 90.48% on the 5th cluster of size 5298 samples. No feature selection technique was employed in this work.

Fotouhi et al. [69] Conducted a comprehensive study on imbalanced cancerous datasets, including breast cancer dataset from the SEER program using 18 data balance oversampling techniques and performed classification using four ML algorithms: MLP, KNN, C4.5 Decision Tree, and RIPPER. The performance of the classifiers has been evaluated using a 5-fold CV and AUC metric. Achieved best AUC value 0.824 on breast cancer dataset using RIPPER+Borderline-SMOTE method. Only AUC has been computed as a performance metric. Data pre-processing or feature selection techniques were not discussed to reduce 133 features into 17.

Abdikenov et al. present a comprehensive prognostic modeling method [70]., yielding a Pareto optimum collection of deep neural networks (DNNs) that provide comparable performance metrics. The ensemble of DNNs is initialized, and its hyperparameters are optimized with the evolutionary method NAGAI. The dataset has been divided into training and testing purposes with a 90:10 ratio, respectively, and the DNN model with a 10-fold CV. A test accuracy of 97.13% and an AUC of 0.9721 were attained with the proposed model.

Simsek et al. [71] A hybrid data mining approach differentiates the essential features for breast cancer survival identification in three periods: 1 year, 5 years, and 10 years. For feature importance, the Least Absolute Shrinkage and Selection Operator (LASSO) and Genetic Algorithm (GA) were utilized, and Random Upsampling (RUS) and SMOTE were employed to address the class imbalance problems in the dataset. Finally, LR and ANN models were applied for classification. By selecting four variables out of 17 using LASSO with RUS, an accuracy of 84.00% and 82.80% was achieved with the LR and ANN models, respectively. Only two ML models were utilized in this study, and the performance was not up to the mark.

Salehi et al. [72] employed MLP ensembles such as MLP stacked generalization (MLPSG) and Mixture of MLP experts (MMLPE) for classifying breast cancer data. The performance of the models has been evaluated using a 10-fold CV and achieved an average accuracy of 81.73%, 84.59%, and 79.39% using MLP, MLPSG, and MMLPE models, respectively, on imbalanced datasets. On a balanced dataset, 88.60%, 83.86%, and 84.32% used MLP, MLPSG, and MMLPE models.

Gupta et al. [73] Performed a comparative analysis in predicting breast cancer using various deep learning algorithms such as ANN, CNN, Restricted Boltzmann Machine, and Deep Autoencoders. SEER dataset with 18 attributes and 163,143 records was chosen after pre-processing, and 5-fold cross-validation was employed to train the deep learning models. Attained best accuracy of 96.80%, F1- score of 96.20%, specificity of 98.00%, MCC score of 93.80%, and an AUC value of 0.968 with the Restricted Boltzmann Machine model. Feature selection techniques were not discussed in this study.

Li et al. [74] Implemented novel ML models for breast cancer prognosis by employing XGBoost and 10-fold CV. Univariate and multivariate COX regression analysis was performed to identify the significant factors from the patients. Further, the model has been validated with clinical data of breast cancer brain metastases (BCBM). It has been concluded that the proposed model was most accurate on 6-month survival patients with an AUC of 0.824.

Wu et al. [75] Evaluated 10 machine learning algorithms for breast cancer diagnosis on the SEER dataset and compared the models with the usual American Joint Committee on Cancer (AJCC) 7th stage. ML algorithms such as NB, LDA, quadratic discriminant analysis (QDA), KNN, SVM, CART, XGB, multivariate adaptive regression splines (MARS), LR, and RF were trained using 9-fold cross-validation. They evaluated their performance by computing accuracy, sensitivity, specificity, F1-score, and AUC. The results concluded that the MARS outperformed other models with an accuracy of 76.4% and an AUC of 0.831. The results are not so accurate and can be improved further. Preprocessing techniques were not discussed—no justification for choosing a 9-fold CV.

Manikandan et al. [76] Proposed a comprehensive ML structure for the classification of SEER breast cancer. Data imputation has been executed using a Random Forest classifier for categorical data and a Random Forest regressor for continuous variables. Variable threshold and PCA approaches were utilized to select significant features and reduce dimensionality. Five machine learning models were trained: Naive Bayes, Decision Tree, AdaBoost, Gradient Boosting, and XGBoost. It was found that DT surpassed the other models, with an accuracy of 98.00% during hold-out and 5-fold cross-validation.

To improve cervical cancer prediction and risk factor assessment, Kolasseri & Venkataramana [77] Conducted a comparative analysis of statistical survival models and machine learning using SEER data. The study evaluates the predictive accuracy and risk factor identification capabilities of Random Survival Forests, Weibull, and Cox proportional hazards models. The results indicate that ML models, especially Random Survival Forests (RSF), surpass conventional statistical techniques in predicting accuracy and identifying significant prognostic factors, highlighting the benefits of machine learning in managing intricate survival data.

A summary of related works on the Surveillance, Epidemiology, and End Results (SEER) Dataset is provided in Table 4.

3.3 Using Ultrasound Images

Radiological imaging techniques like mammography, ultrasound (US), and magnetic resonance imaging (MRI) are frequently used to screen for and diagnose breast lesions. While mammography is considered the gold standard method for breast cancer assessment, its imaging capabilities are limited by the dense breast; as a result, US and MRI are regarded as supplemental breast-screening procedures [78]. Medical ultrasound (US) scans are extensively utilized owing to their advantages, including cost-effectiveness, rapidity, and non-invasiveness, and have become a prevalent tool for detecting breast cancer [79, 80]. Finding essential features in an image is a complex and time-consuming process. Furthermore, it can be challenging to optimize traditional CAD's overall performance.

Abdel-Nasser et al. [81] Focused on artifacts and speckle noise that appears in breast ultrasound images for classification purposes using texture analysis. The dataset, comprising 28 benign and 31 malignant, has been collected from the Addenbrooke's Hospital, Cambridge. The low-resolution images were converted to high resolution using super-resolution methods, and the features from the segmented images were extracted using five texture analysis techniques. The features have been fed to the Random Forest classifier to classify the breast ultrasound images. The proposed CAD system achieved an AUC of 0.989 using the features extracted from the histogram of oriented gradients.

Qi et al. [82] Proposed convolutional neural networks for automatically diagnosing breast cancer from ultrasound images by constructing a vast, annotated database of 8045 images by contacting professional surgeons. The work has been divided into recognizing the malignant tumors and identifying solid nodules in the photos. Region enhancement technique (REM) has been employed to improve the performance of the proposed model. Cross-training was utilized to train the deep-learning model. The entire dataset was divided into 4/6, 1/6, and 1/6 parts for training, validation, and testing. Finally, they achieved an accuracy of 94.48% and an AUC of 0.982 on malignant detection with REM. Meanwhile, 90.13% accuracy was achieved in solid nodule recognition, and an AUC of 0.928 was achieved.

Moon et al. [83] Established a computer-aided detection system for detecting tumors in breast ultrasound using a 3-D CNN algorithm. The research utilized the sliding window technique to split the breast ultrasound, ensemble, and focal loss methods to address the dataset imbalance issues, and 3-dimensional convolutional neural networks to classify breast ultrasound images. The dataset used in this work has been collected from the Seol National University Hospital. The proposed model has achieved a sensitivity of 95.3%. The main limitation is that the number of false positives is very high, and only one performance measure has been considered in this work.

Pavithra et al. [84] Developed a computer-aided diagnosis system for classifying breast ultrasound images, whether regular, benign, or malignant, using the KNN, DT, and Random Forest (RF) algorithms. During the pre-processing, the reducing Anisotropic Diffusion method was utilized to remove the speckle noise, and GLCM was employed to extract the texture feature and train the ML algorithms. RF algorithms predicted with better accuracy than the remaining two algorithms.

Wang et al. [85]Collected 316 ultrasound breast lesion images (181 benign and 135 malignant) from Jeonbuk National University Hospital. The Multiview CNN using InceptionV3 has been utilized for feature extraction, and the proposed method has been trained using a 5-fold cross-validation method. The proposed method has achieved a Sensitivity of 80.60%, a Specificity of 87.60%, and an AUC of 0.9468. Claimed that the proposed mode AUC is >10% when compared with the methods utilizing principal component analysis (PCA) and histogram of oriented gradients (HOG).

Shia and Chen [86] Employed transfer learning for classifying the breast ultrasound images, whether malignant or benign. The image dataset comprises 2099 ultrasound

Table 4 Summary of related works on surveillance, epidemiology, and end results (SEER) dataset

Author & Year	Feature Selection technique	No. of features	ML/DL Algorithm	Max. accuracy achieved	Limitations
Shukla et al. [69]	–	29	Multilayer Perceptron	90.48%	No feature selection technique was employed in this work.
Fotouhi et al. [70]	–	17	RIPPER + Borderline-SMOTE	–	Only AUC (0.824) has been computed as a performance metric.
Abdikenov et al. [71]	–	19	DNN	97.13%	The study can be improvised by employing feature selection methods.
Simsek et al. [72]	LASSO with RUS	4	LR	84.00%	Only two ML models were utilized in this study, and the performance was not up to the mark.
Salehi et al. [73]	–	35	MLP	88.60%	The research has not delineated the technique for feature selection. And the predictive outcomes are insufficiently significant.
Gupta et al. [74]	–	18	Restricted Boltzmann Machine model	96.80%	Feature selection techniques were not discussed in this study.
Li et al. [75]	Univariate and Multivariate COX analysis	18	XGBoost	–	Only AUC (0.824) has been computed as a performance metric. The prediction outcomes are not good enough.
Wu et al. [76]	Boruta Package	14	MARS	76.40%	The results are not so accurate and can be improved further. Pre-processing techniques were not discussed—no justification for choosing a 9-fold CV.
Manikandan et al. [77]	Variable threshold and PCA	13	Decision Tree	98.00%	The authors have employed standard ML techniques in this work.
Kolasseri & Venkataramana [78]	–	11	Random Survival Forest	–	The error rate of 0.0019 was achieved by the RSF method. The heterogeneity of data, encompassing transformations and normalizations, is a considerable obstacle to the extensive application of ML methods.

images from 543 individuals collected in this study. A trained deep residual network (ResNet-101) was utilized to extract the features from the image dataset, and linear support vector machines were employed to classify the images. They've achieved 94.34% of sensitivity, 93.22% of specificity, and an AUC value of 0.938. The proposed method outperformed physician diagnosis, which achieved an AUC value of 0.603. Advanced deep-learning techniques can further improve the performance of the model.

Shia et al. [87] Collected 677 breast ultrasound images from Changhua Christian Hospital, out of which 418 are malignant and 370 are benign. Image features such as corners, edges, and shapes were extracted using pyramid histogram-oriented gradients (PHOG). Significant features have been selected using the correlation-based feature selection (CFS) method. Sequential minimal optimization with locally weighted learning has been employed to classify the breast images. The model achieved a Sensitivity of 81.64%, Specificity of 87.76%, and AUC of 0.847.

Ayana et al. [88] Proposed a novel multistage transfer learning for classifying breast ultrasound images. They've collected 20,400 microscopic cancer cell images and 600 ultrasound images from various databases. ImageNet has been applied to convert microscopic images like US images. The pre-trained models, such as EfficientNetB2, InceptionV3, and ResNet50, were used with Stochastic gradient descent, Adam, and Adagrad optimizers by splitting the entire dataset of images into 70:20:10 ratio for training, validation, and testing. ResNet50 has achieved an average testing accuracy of 98% with 5-fold cross-validation.

Chen et al. [89] Explored GoogLeNet, a CNN model, to classify breast ultrasound (BUS) images. A dataset includes 880 BUS images with 103 regular, 210 benign, and 467 malignant tumors. The imbalance issues were resolved by employing a random upsampling method. The speckle noise from the images has been eliminated using median filtering. Experts have manually annotated the entire preprocessed dataset. The dataset has been split into a 60:20:20 ratio for training, verification, and testing purposes. Finally, the image classification was performed using Google, with an average accuracy of 96.37%.

Alotaibi et al. [90] Projected CNN and image fusion methods for breast cancer classification using ultrasound images. They identified three US datasets (BUSI, dataset B, and KAIMRC), public and private, with a total of 6635 images. Various pre-processing steps were incorporated, such as cropping, denoising, highlighting interested regions of images, and RGB fusion. Block matching 3D filtering has been utilized to denoise the images. The dataset was divided into 80:20 ratios for training and testing, and the VGG19 CNN model was trained for extracting features and classification using 5-fold cross-validation. A mean accuracy of 87.80% has been achieved on the BUSI dataset.

The summary of related works on Ultrasound images has been provided in Table 5, which includes the Author's name and year of publication, the number of ultrasound images considered in the work, the size of the photos, preprocessing techniques utilized, the ML/DL algorithm employed, best performance measures achieved by them, and the limitations of the work.

3.4 Using Mammogram Images

Abdel-Nasser et al. [91] Analyzed mammogram images for abnormality and breast density based on the features extracted from the uniform local directional pattern (ULDP). This work considered two mammogram datasets, mini-MIAS with 322 pictures and INbreast with 410 images. The images were segmented according to region of interest, and ULDP features were applied to the linear and non-linear SVM classifier. Achieved an AUC value of 0.9324 using LSVM on the mini-MIAS dataset and 0.9930 using LSVM on the INbreast dataset.

Sadad et al. [92] Cogitated 109 images from MIAS and 72 images from DDSM datasets. Image cropping has been performed to extract ROIs from the mammograms. Combined fuzzy C-means and region-growing methods (FCMRG) were utilized for segmentation. Local binary pattern GLCM (LBP-GLCM) and local phase quantization (LPQ) methods were employed to extract the features from the segmented images. The minimum redundancy maximum relevance algorithm has been used to select significant features and train six ML models: DT, LDA, LR, SVM, KNN, and Ensemble with 10-fold cross-validation. Achieved the highest accuracy of 98.2% using DT on the MIAS dataset.

Shen et al. [93] Developed a deep learning technique for accurate prediction of cancer from breast mammograms. This work collected the mammogram images from a curated breast imaging subset-digital database for screening mammography (CBIS-DDSM) and INbreast datasets with five target classes: background, benign calcification, malignant calcification, benign mass, and malignant mass. An AUC value of 0.88 was achieved on the CBIS-DDSM dataset and 0.91 on the INbreast dataset.

Vijayarajeswari et al. [94] Collected three types of mammogram images (malignant, benign, and standard) from the Mammogram Image Analysis Society (MIAS) database. The noise and redundant information related to the pectoral muscle have been removed using the maximum likelihood estimation method during the preprocessing stage. A 2-dimensional Hough transform has been utilized to extract the feature from the preprocessed images. Thereafter, extracted features were applied to the SVM classifier to

Table 5 Summary of related works on ultrasound images

Author & Year	No. of Ultrasound Images studied	Size of the images (pixels)	Preprocessing Techniques	ML/DL Algorithm	Performance measures	Limitations
Abdel-Nasser et al. [82]	59 (Collected from Addenbrooke's Hospital, Cambridge)	Not specified	Gray level co-occurrence matrix, Local binary pattern, Phase concurrency based local binary pattern, Histogram of oriented gradients and pattern lacunarity spectrum	Random Forest	AUC − 0.989	The dataset is too small, with only 59 images. The resolution of the pictures is inferior.
Qi et al. [83]	8145	299 × 299	Region enhancement method, Multi-input prediction	Convolutional Neural Networks	Accuracy − 94.48% Sensitivity − 95.65% Specificity − 96.66% AUC − 0.992	The dataset is imbalanced with 5386 non-malignant and 2759 malignant images.
Moon et al. [84]	246 (Collected from Seol National University Hospital)	30 × 30 × 30	Sliding Window Method, Candidates aggregation algorithm	3-D Convolutional Neural Networks	Sensitivity − 95.30%	Only one performance measure has been computed.
Pavithra et al. [85]	780	1280 × 1024	Speckle Speckle-reducing anisotropic Diffusion, active contour-based segmentation, GLCM	KNN, DT, RF	Accuracy − 88.00%	Preprocessing and validation method techniques were not discussed.
Wang et al. [86]	316 (Collected from Jeonbuk National University Hospital)	299 × 299 × 3	Not mentioned	InceptionV3 CNN	Sensitivity − 80.60% Specificity − 87.60% AUC − 0.9468	No preprocessing steps were discussed. 5-fold cross-validation used.
Shia and Chen [87]	2099	224 × 224	Random affine geometric transformation	ResNet-101, Linear SVM	Sensitivity − 94.34% Specificity − 93.22%	Advanced deep-learning techniques can further improve the performance of the model.
Shia et al. [88]	677 (Collected from Changhua Christian Hospital)	Not specified	Histogram-oriented gradient descriptors, Correlation-based feature selection	Sequential minimal optimization with local weight learning	Sensitivity − 81.64% Specificity − 87.76% AUC – 0.847	The performance of the proposed model can be improved further.
Ayana et al. [89]	21,000	75 × 75	ImageNet	EfficientNetB2, InceptionV3, ResNet50	Avg. test accuracy − 98.00%	No specific justification was provided for using 5-fold cross-validation.
Chen et al. [90]	880	768 × 1024	Random upsampling, Median filtering	GoogLeNet	Accuracy − 96.37%	Optimization or hyperparameter tuning methods were not discussed in the work.
Alotaibi et al. [91]	6635	Not specified	Cropping, denoising, ROI highlighting, RGB fusion	VGG19 CNN model	Mean accuracy − 87.80%	Unable to perform cropping and highlighting ROI on one of the local datasets.

classify the images. An accuracy of 94% has been achieved with the proposed model. Huang et al. [95] Considered 410 mammogram images from the INbreast dataset and performed various preprocessing steps, including data augmentation, contrast limited adaptive histogram equalization (CLAHE), and normalization. After data augmentation, they obtained 7632 images with eight target classes. Then, the dataset was split into 80:20 ratio for training and testing sets, respectively, and three CNN models, AlexNet, DenseNet, and ShuffleNet, using 5-fold cross-validation. The best accuracy of 97.62% was achieved by the DenseNet model.

Muduli et al. [96] Developed a CAD model for breast cancer prediction using fast discrete curvelet transform (FDCT) and modified particle swarm optimization-based extreme learning machine (MODPSO - ELM). The images from three standard mammogram datasets, MIAS with 326 images, DDSM (digital database for screening mammography) with 1500 images, and INbreast with 410 photos, were considered in this work. The ROIs were extracted by manual cropping. The features were extracted using FDCT with

a wrapping method and two feature reduction algorithms, PCA and linear discriminant analysis (LDA), to reduce redundant or unwanted features. Attained best accuracies of 100%, 98.94%, and 98.76% on MIAS, DDSM, and INbreast datasets, respectively.

Maqsood et al. [97] presented a transferable texture CNN (TTCNN) method for early breast cancer diagnosis using digital mammograms. They've collected images from three standard datasets: DDSM, INbreast, and MIAS. CLAHE algorithms were used for contrast enhancement as part of preprocessing. Various pre-trained deep CNN models like InceptionResNet-V2, Inception-V3, GoogLeNet, VGG-16, VGG-19, ResNet-18, ResNet-50, and ResNet-101 were utilized for feature extraction. Convolutional sparse image decomposition (SID) and entropy-controlled firefly (ECF) approaches were employed for feature fusion and selection. An accuracy of 99.08% was achieved on the DDSM dataset, 96.82% on the INbreast dataset, and 96.57% on the MIAS dataset using the proposed approach.

Pattanaik et al. [98] Introduced a hybrid model called DenseNet121 with an extreme learning machine for breast mammogram classification. The DDSM dataset has been considered in this work, and the number of 11,000 images has increased by performing data augmentation. The image database has been split into 70% for training and 30% for testing. DenseNet121 + ELM was trained by adjusting its weights using the AdaGrad optimization method. Achieved testing accuracy of 99.14%, specificity of 99.47%, and sensitivity of 99.94% with the proposed method.

Jabeen et al. [99] Utilized INbreast and curated breast imaging subset of DDSM (CBIS-DDSM) datasets for the experimentation of breast cancer classification. In phase 1, the steps of contrast enhancement to improve the quality of the images and data augmentation to increase the image quantity were implemented. Haze-reduction local-global (HRLG) method was deployed to enhance the photos. Pre-trained EfficientNet-B0 has been employed to train original and enhanced image datasets by extracting the features from the average pooling layer. Equilibrium-Jaya controlled Regula Falsi (EJCRF) method was used to select the appropriate features. The dataset with extracted features was split into a 50:50 ratio for training and testing purposes. Numerous ML and DL algorithms such as ensemble subspace KNN, fine KNN, cubic SVM, medium Gaussian SVM, medium neural networks, wide neural networks, and weighted KNN, QSVM, N^3 and BN^2 were trained with 10-fold cross-validation. Achieved accuracies of 95.4% and 99.7% on CBIS-DDSM and INbreast datasets, respectively.

Melekoodappattu et al. [100] Implemented an improved CNN model for effective breast mammogram categorization on MIAS and DDSM repositories. A multi-scale local binary pattern (MLBP) method was used to extract the features. A Gabor filter was utilized to detect the breast abnormalities. The study employed linear discriminant analysis (LDA) and uniform manifold approximation and projection (UMAP) methods to reduce the dimensionality of the dataset. Ensemble models have been created by combining CNN with ELM, RBFNN, SVM, KNN, and WNN and trained based on the selected texture features. Achieved testing accuracy of 98% on the MIAS dataset and 97.9% on the DDSM dataset.

Chugh et al. [101] Utilized two principal methodologies for the diagnosis of breast cancer. The initial approach involves deep feature extraction using a machine learning classifier head, employing Deep Convolutional Neural Network models such as VGG16, VGG19, ResNet50, and ResNet152 as feature extractors, then training traditional machine learning classifiers like Random Forest and XGBoost. The alternative approach involves Deep Learning feature extraction with a neural network classifier, employing models such as MobileNet, VGG16, VGG19, ResNet50, ResNet152, and DenseNet169 for feature extraction and classification. In the initial methodology, Random Forest (RF) and XG Boost (XGB) classifiers attained 100% accuracy on the training set, whilst Support Vector Machine (SVM) and XGB demonstrated 95% accuracy on the test dataset. The second method, employing Mobile Net, ResNet50, and Dense Net 169, attained an accuracy of 97% for both training and testing datasets.

Liu et al. [102] Developed an improved VGG16 model for breast cancer classification using mammogram images. A dataset consisting of 7841 mammograms has been collected from Jinan University's hospital. Data augmentation was performed to increase the number of pictures to 12fold. The dataset class imbalance problem was addressed using the focal loss function method. The dataset has been split into a 70:30 ratio for training and testing, respectively. Adam optimization method was applied to tune the models' parameters automatically. The proposed model was trained using mini-batches with 32 images. Achieved an average accuracy of 96.94% with the proposed algorithm. The summary of related works on mammogram images has been provided in Table 6.

3.5 Using BreakHis Dataset

The BreakHis dataset is a well-known dataset used for breast cancer histopathological image classification. Microscope biopsy images of benign and malignant breast cancers were gathered from January to December 2014 as part of a clinical trial and are included in the BreakHis dataset. To participate in the study, patients at the Hospital de Câncer de Barretos in Brazil had to consent. The research was authorized by an institutional evaluation board, ensuring all

Table 6 Summary of related works on mammogram images

Author & Year	The dataset and number of Mammogram Images studied	Size of the image (pixels)	Preprocessing and feature extraction techniques	ML/DL algorithm	Performance measures	Limitations
Abdel-Nasser et al. [92]	Mini-MIAS (322) INbreast (410) Total-732	32 × 32, 64 × 64, 75 × 75 150 × 150	Uniform local directional pattern	Support Vector Machines	AUC − 0.9930	Removal of unwanted information like pectoral muscle and speckle noise not addressed.
Sadad et al. [93]	MIAS (109) DDSM (72) Total-181	224 × 224	Cropping, FCMRG, LBP-GLCM, LPQ, mRMR	DT, LDA, LR, SVM, KNN, Ensemble	Sensitivity − 100.0% Specificity − 97.00% Accuracy − 98.20%	Worked with a smaller number of images.
Shen et al. [94]	CBIS-DDSM INbreast Total-2478	224 × 224	Not discussed.	VGG16 and Resnet50		The preprocessing techniques implemented were not disclosed.
Vijayarajeswari et al. [95]	MIAS (95)	1024 × 1024	Hough transform	Support Vector Machines	Accuracy − 94.00%	The dataset considered is too small in this work.
Huang et al. [96]	INbreast (410) After DA-7632	224 × 224 227 × 227	CLAHE, Data augmentation, Image normalization	AlexNet, DenseNet, ShuffleNet	Accuracy − 99.72%	Dataset class imbalance steps were not addressed.
Muduli et al. [97]	MIAS (326), INbreast (410), DDSM (1500) Total-2236	128 × 128	FDCT with wrapping, PCA, LDA	MODPSO-ELM	Accuracy − 100% AUC − 1.00	During preprocessing, manual cropping of images takes more time.
Maqsood et al. [98]	MIAS (326), INbreast (410), DDSM (1500) Total-2236	224 × 224	CLAHE, SID, ECF	InceptionResNet-V2, Inception-V3, GoogLeNet, VGG-16, VGG-19, ResNet-18, ResNet-50, and ResNet-101	Accuracy − 99.08% Sensitivity − 99.19% Specificity − 98.96% AUC − 0.9908	The assessment has been performed on individual datasets.
Pattanaik et al. [99]	DDSM (1500) After DA-11,000	224 × 224	Data augmentation, resizing, and normalization	DenseNet121 + ELM	Accuracy − 99.14% Sensitivity − 99.94% Specificity − 99.37%	Dataset imbalance and noise removal issues were not addressed.
Jabeen et al. [100]	CBIS-DDSM After DA-1910	380 × 380	HRLG, EJCRF	EfficientNet-B0, MGSVM, ESKNN, QSVM, F-KNN, LSVM, Bi-NN, Tri-NN, CSVM	Accuracy − 99.70%	Manual hyperparameter tuning is a time-consuming and infective method.
Melekoodappattu et al. [101]	MIAS (326) and DDSM (1500)	-	LBP	Ensemble CNN	Accuracy − 98.00% Specificity − 97.80%	This work did not consider images, and image enhancement methods were not discussed.
Chugh et al. [102]	CBIS-DDSM	299 × 299	Min-Max normalization, CLAHE, Padding, and Centre-crop, Data augmentation	DCNN, RF and XGB	Accuracy − 100%	The effectiveness of deep learning models often hinges on the availability of large datasets for training.
Liu et al. [103]	Collected from Hospital of Jinan University (7841)	224 × 224	Data augmentation, focal loss function	VGG16 CNN	Avg. accuracy − 96.94%	Noise removal steps were not addressed in this work.

ethical guidelines were followed. Samples were taken from biopsy slices of breast tissue stained with haematoxylin and eosin (H&E). A pathologist from the Barretos Cancer Hospital used a light microscope to analyze these slides. The pictures depict several kinds of tumors frequently categorized in medical diagnostics. A uniform methodology was used during the imaging process. A microscope was used to scan each slide, and pictures were taken at predetermined magnifications. Following the scanning process, the images were saved and categorized. A digital camera system and an Olympus BX-50 system with a relay lens and a 3.34 magnification were used for scanning. The objective lenses were set at 4x, 20x, 40x, and 60x, and the images were taken at three distinct magnifications: 40x, 100x, and 400x. The

dataset consists of 7909 breast images, with 2480 belonging to benign (31.35%) and 5429 to malignant (68.64%). With more than 7,000 photos, the dataset offers significant information for machine learning model testing and training. Since the images are usually saved in standard file types (such as JPEG or PNG), they can be used with various image processing and machine-learning tools. It acts as a standard by which to measure how well various categorization algorithms perform in the area of medical image analysis.

Spanhol et al. [103] proposed CNN-based histopathological image classification for diagnosing breast cancer. They've considered the extracted breast image patches from the BreakHis dataset. The dataset has been split into training (70%) and testing (30%) datasets. The CNN model was trained on the extracted dataset and achieved an accuracy of 89.6% on the image level with the 40x magnification factor. The results obtained were compared with the patient level, and it was concluded that the proposed model provided good results for clinical diagnosis of breast cancer at the early stage.

Spanhol et al. [104] conducted a study on breast cancer (BC) diagnosis via deep learning methodologies, particularly examining the effectiveness of deep convolution activation features (DeCAF) as a substitute for conventional hand-crafted features and task-specific convolutional neural networks. The study underscores the constraints of CNNs regarding complexity and training duration while illustrating that DeCAF features obtained from a pre-trained CNN can expedite the creation of precise BC classification systems. The experimental findings demonstrate that DeCAF features surpass conventional approaches and sometimes compete with task-specific CNNs, facilitating progress in automated cancer identification. An F1-score of 93.50% (patient level) has been attained at a 4x magnification level.

Nahid et al. [105] employed many strategies for breast cancer image classification, primarily utilizing Deep Neural Network (DNN) techniques, notably Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models. It utilizes clustering techniques, such as K-Means and Mean-Shift, to extract structural and statistical information from the images. The classification process utilizes SoftMax and Support Vector Machine (SVM) layers to make decisions. The research achieved an accuracy of 91.00% on the 200x dataset, with a peak precision of 96.00% on the 40x dataset and the best F-measure on both the 40x and 100x datasets.

Kumar et al. [106] utilized a Convolutional Neural Network (CNN) for feature extraction and classification of breast cancer images from the BreakHis dataset. It employs a deep learning architecture to improve picture identification abilities, primarily focusing on categorizing benign and malignant tumors. The training procedure employs Stochastic Gradient Descent (SGD) and back-propagation methods, utilizing a categorical cross-entropy objective function. Cross-validation is conducted using a 10-fold approach to guarantee thorough model assessment and attained an accuracy of 90.00% utilizing the proposed methodology.

Deniz et al. [107] utilized transfer learning and deep feature extraction techniques to identify breast cancer from histopathology pictures. Pre-trained CNN models, namely AlexNet and VGG16, are employed for feature extraction, utilizing the fc6 and fc7 layers to derive feature vectors. An SVM classifier categorizes the collected features into benign and malignant classifications. The optimized AlexNet model achieved the highest average accuracy across all magnification levels, attaining a maximum score of 93.57 for the 40X magnification factor in Fold 3. The highest accuracy for the 100X magnification factor was 89.34 in Fold 1, but the 200X magnification factor achieved a maximum accuracy of 93.02 in Fold 3.

Gour et al. [108] introduced a novel residual learning-based convolutional neural network model, ResHist, specifically developed for the automated categorization of breast cancer histopathology pictures. The research outlines the creation of a 152-layer deep residual learning architecture that can differentiate between benign and malignant tissues in biopsy samples, attaining significant results with an accuracy of 92.52% and an F1-score of 93.45%, bolstered by a specifically designed data augmentation technique. The results demonstrate that the ResHist model surpasses current approaches and pre-trained networks in breast cancer diagnosis.

Yamalome et al. [109] propose a Convolutional Neural Network (CNN) methodology that integrates transfer learning, data augmentation, and comprehensive image analysis to improve diagnostic precision and prevention in classifying breast cancer histopathology images. The suggested method utilizes transfer learning, data augmentation, and whole-picture training instead of traditional patch training to improve classification accuracy. The proposed model attains an average accuracy of 91% at the image level and 95% in patient assessments, signifying a significant enhancement over current methodologies and offering a reliable automated instrument for breast cancer diagnosis.

A novel CNN architecture created especially for categorizing breast cancer histopathology images is used by Uppada et al. [110] It extracts characteristics from histopathology pictures using Transfer Learning and a pre-trained DenseNet-201 model. Techniques for data augmentation are used to increase the quantity and diversity of the training dataset. An accuracy of 99.7%, precision of 93%, sensitivity of 97%, F1-Score of 95%, and ROC-AUC of 0.88 were among the model's performance metrics.

Wang et al. [111] introduced a breast cancer picture classification approach integrating deep learning and transfer learning methodologies. It employs the DenseNet architecture of deep neural networks, including attention mechanisms, to improve the model's efficacy. The experimental configuration encompasses data preprocessing and augmentation to alleviate overfitting and enhance classification accuracy. The model exhibited improved performance in binary classification tasks on the BreakHis dataset at different magnification levels, achieving accuracies between 80.1% and 84.3% from 40x to 400x magnification.

Guo et al. [112] proposed a novel breast cancer image classification model (BCICM) that utilizes texture feature parameter extraction (TFPE) to convert images into matrices for analysis. The images were collected from the BreakHis database, and image processing steps were performed to make them into 40x, 100x, 200x, and 400x pixel sizes. The texture features were extracted using multi-scale texture feature extraction methods (GLCM and Tamura coarseness), and classification was performed using an unsupervised dynamic learning algorithm by splitting the data into 80:20 ratios for training and testing. The proposed method has achieved an accuracy of 91.50% and an AUC value of 0.915.

Balasubramanian et al. [113] introduced a novel methodology employing ensemble deep-learning algorithms to categorize breast tumor subtypes and evaluate invasiveness via histopathology images. The research utilizes convolutional neural networks (CNNs) such as VGG16 and ResNet, implementing an innovative image patching technique on recognized datasets (BACH and BreakHis) to improve diagnostic precision. Image patching was implemented as a preprocessing method, enabling concentrated examination of particular areas within high-resolution histopathology images. Patch classification accuracy on the BACH dataset was 95.31%, while on the BreakHis dataset was 98.43%. A summary of related works on the BreakHis dataset is provided in Table 7.

4 Discussion

Table 3 summarizes related works concerning the Wisconsin Breast Cancer Dataset. The document details the author and publication year, the feature selection methodology, the number of features selected, the machine learning or deep learning (ML/DL) algorithm applied, the highest accuracy attained, and the study's limitations. Diverse feature selection methodologies, including the Binary Bat Algorithm, Gain Ratio, Chi-square, ReliefF, Recursive Feature Elimination, and Gravitational Search Algorithm, have been employed. A variety of ML/DL techniques have been utilized, including OGNN, Extreme Learning Machine, Shallow ANN, Gradient Boosting, RF, SVM, Stacking, Gaussian Naive Bayes, Dense CNN, and Convolutional Neural Networks. The documented maximum accuracies range from 93.53 to 99.89%. Common limitations encompass the absence of data pre-processing procedures, feature selection methodologies, and the resolution of class imbalance within the dataset. Several research studies also failed to elaborate on hyperparameter tuning or feature selection techniques comprehensively. The table shows the variety of methods employed to analyze the Wisconsin Breast Cancer Dataset and the possibility of further enhancing accuracy and robustness by resolving the limitations found.

Table 4 encapsulates pertinent studies concerning the Surveillance, Epidemiology, and End Results (SEER) Dataset. Diverse feature selection methodologies, including LASSO with RUS, Univariate and Multivariate COX analyses, the Boruta Package, variable thresholding, and Principal Component Analysis (PCA), have been employed. A variety of ML/DL algorithms have been utilized, including Multilayer Perceptron, RIPPER combined with Borderline-SMOTE, Deep Neural Network, Logistic Regression, MLP, Restricted Boltzmann Machine, XGBoost, Multivariate Adaptive Regression Splines, Decision Tree, and Random Survival Forest. The recorded accuracies range from 76.40 to 98.00%. Common shortcomings encompass the absence of feature selection methodologies, a restricted variety of machine learning models employed, inadequately substantial predictive findings, erroneous outcomes, insufficient discourse about pre-processing procedures, and the rationale for selecting 10-fold cross-validation. The table underscores the various methodologies employed in analyzing the SEER Dataset and the opportunity for enhanced accuracy and robustness by rectifying the observed shortcomings.

Table 5 summarizes related works on breast cancer tumor classification using breast ultrasound images. The number of pictures studied varies significantly across studies, ranging from 59 to 8145. Image sizes also differ, with some studies using smaller images (30×30 pixels) and others using larger images (299×299 pixels). Standard preprocessing techniques include region enhancement, multi-input prediction, speckle anisotropic diffusion, active contour-based segmentation, and geometric transformations. Several ML/DL algorithms have been applied, including Random Forest, CNNs, KNN, DT, RF, Inception V3 CNN, ResNet-101, Linear SVM, and Sequential minimal optimization. The primary performance measures reported include AUC, accuracy, sensitivity, and specificity. The accuracies attained on the breast ultrasound dataset are between 87.80% and 98.00%. Typical limitations include small dataset size, imbalanced datasets, poor image resolution, limited performance measures, lack of discussion on preprocessing and

Table 7 Summary of related works on breakhis dataset

Author & Year	Dataset and No. of Break-His Images studied	Size of the input image	Preprocessing and feature extraction techniques	ML/DL algorithm	Performance measures	Limitations
Spanhol et al. [104]	7909	32 × 32 64 × 64	Image fusion	AlexNet	Accuracy −89.60%	While the work achieved an accuracy of 89.60%, it did not address dataset imbalance issues or perform hyperparameter tuning, essential factors in model performance.
Spanhol et al. [105]	7909	700 × 460	Not discussed	CNN	F1-score −93.50% (patient level) F1-score −92.9% (image-level)	primarily focused on the F1-score of patient and image levels as performance metrics, a common approach in medical image classification due to its balance between precision and recall.
Nahid et al. [106]	7909	760 × 360	Not specified	CNN, LSTM, RNN SoftMax, and SVM	Accuracy −91.00% Precision −96.00%	Pre-processing techniques were not discussed in the work. Accuracy can be improved further.
Kumar et al. [107]	7909	32 × 32 64 × 64	Feature extraction using CNN	CNN	Accuracy −90.00%	While the work has achieved a commendable accuracy of 90.00%, there is still room for improvement. With continued research and development, we can strive to reach as close to 100% accuracy as possible, offering hope for more effective breast cancer classification.
Deniz et al. [108]	7909	227 × 227 224 × 224	Not specified.	AlexNet, VGG16, and SVM	Accuracy −93.57%	The complexity of tuning many parameters in CNN adds to the system's intricacy.
Gour et al. [109]	7909	224 × 224	Data augmentation	Residual Learning-based CNN with Adam optimization	Accuracy −92.52% F1-Score −93.45%	The authors claimed to validate the proposed model on large datasets to improve the accuracy.
Yamlome et al. [110]	7909	700 × 460	Data augmentation	CNN	Accuracy −91.00% (image-level) Accuracy −95.00% (patient level)	Risk of overfitting due to the data augmentation. The performance may not be satisfactory from a clinical perspective.
Uppada et al. [111]	7909	224 × 224	Data augmentation	DenseNet-201	Accuracy −99.75%	This work did not address the dataset imbalance and overfitting due to augmentation issues.
Wang et al. [112]	7909	40x, 100x, 200x, and 400x pixels	Data augmentation	DenseNet	Accuracy −84.00%	The model focuses on binary classification without grading or subtyping of breast cancer.
Guo et al. [113]	Not mentioned	40x, 100x, 200x, and 400x pixels	GLCM and Tamura coarseness	Unsupervised dynamic learning mechanism (UDLM)	Accuracy −91.50% AUC value −0.915	The obtained accuracy was not remarkable. It can be improved further.
Balasubramanian et al. [114]	7909	700 × 460	Image patching and segmentation	Ensemble model (VGG-Net 16 and ResNet 50)	Accuracy – 98.43%	Data preprocessing steps were not discussed, and Dataset imbalance issues were not addressed.

validation techniques, and the potential for further improvement through advanced deep learning techniques.

Table 6 presents a comparison of various studies on breast cancer classification using mammogram images. The datasets used in each study are MIAS, DDSM, and INbreast, with pictures ranging between 95 and 7841. Preprocessing techniques, such as cropping, resizing, noise reduction (e.g., CLAHE), feature extraction (e.g., LBP-GLCM, LPQ), and data augmentation, are applied to the images. Techniques used to extract relevant features from the pre-processed images, such as Local Binary Patterns (LBP), Gray-Level Co-occurrence Matrix (GLCM), and

Local Phase Quantization (LPQ). The classification algorithms employed in each study include SVM, Ensemble methods, and CNNs like AlexNet, DenseNet, and ResNet, among others. The performance metrics used to evaluate the classification models are accuracy, sensitivity, specificity, and AUC. The accuracies reported on breast mammograms are in the range of 94.00% and 100%. Each study's limitations include small dataset size, class imbalance, and the need for further evaluation of individual images, highlighting the existing gaps in research that need to be addressed.

Table 7 summarizes several studies on breast cancer classification using the BreakHis dataset. All studies utilize the BreakHis dataset, an extensive collection of breast histopathology images. The size of the images varies between studies, ranging from 32 × 32 pixels to 700 × 460 pixels. Some studies employ image fusion techniques, while others do not explicitly mention preprocessing steps. One study utilizes a CNN for feature extraction. Various models are used, including AlexNet, CNNs, LSTMs, RNNs, SVMs, and Residual Learning-based CNNs. The accuracy range attained on the BreakHis dataset is from 84.00 to 99.75%. Some studies acknowledge the issue of class imbalance in the dataset, lack of pre-processing, complexity in tuning a large number of parameters of the DL models and overfitting due to data augmentation.

The incidence of breast cancer is escalating at a concerning rate globally.

Mortality rates in affluent nations are markedly elevated and are increasing across all global regions. Timely identification and diagnosis of breast cancer are essential for enhancing long-term survival rates. Medical Image Analysis (MIA) utilizing Computer-Aided Diagnosis (CAD) has demonstrated the highest efficacy for the early detection and treatment of breast cancer (BC). The application of deep learning models for breast cancer diagnosis significantly aids physicians in reaching a conclusive verdict. Deep learning does not necessitate explicit feature extraction methods, as it autonomously learns low-level, mid-level, and high-level features. A higher number of papers have been released about the severity of BC. Consequently, aggregating all research works employing deep learning methodologies into a singular piece is a formidable challenge. This work offers a comprehensive method that encompasses the existing breast cancer databases, segmentation techniques, and recent insights on the classification and detection of breast cancer.

This work indicates that mammography and histopathology (BreakHis dataset) are prevalent imaging modalities utilized to assess breast cancer, and the literature also demonstrates that mammography is an efficacious method for the early diagnosis of BC with 100% accuracy. Nonetheless, mammogram pictures exhibit limited contrast; hence, various noise reduction and contrast enhancement techniques are utilized during the pre-processing step. Research indicates that the rates of false positives and negatives in classification diminish when preprocessing and data augmentation approaches are used for breast cancer images. The majority of mammography research concentrates on the detection of calcifications and masses. Researchers may employ CNN-based methods to enhance density classification and identify and classify asymmetry areas with limited associated publications. The majority of deep learning algorithms are trained on publicly accessible datasets. The dimensions and standards of these datasets significantly influence the effectiveness of these systems. Training deep learning models on these datasets may enhance their performance. Moreover, our analysis reveals that improving deep learning models for constrained medical imaging datasets remains an unresolved research topic. Nevertheless, most writers have employed transfer learning, data augmentation, and regularization techniques to mitigate overfitting and enhance model performance for constrained datasets. When combined with hand-engineered feature extraction techniques, deep learning yielded superior results.

4.1 Datasets for Breast Cancerous Tumor Classification

Table 8 provides the availability of various datasets used for breast cancerous tumor classification with the web links.

4.2 Limitations of AI Models in Detecting Rare or Ambiguous Breast Cancer Cases

AI models are frequently trained on extensive datasets where invasive ductal carcinoma (IDC) predominates, but infrequent subcategories (for example, medullary, mucinous, or tubular carcinomas) are inadequately represented. Consequently, AI models often overfit to prevalent traits and suffer difficulties in generalizing when faced with infrequent histological subtypes. The production of synthetic data through generative adversarial networks (GANs) and the application of re-weighted loss functions to penalize incorrect classification of rare tumor kinds will assist in resolving this issue.

In women with high breast density, AI models might not be able to differentiate small tumors from typical fibro glandular structures. They may encounter challenges in identifying the lesions due to the low-contrast photos. The multimodal AI technique combining MRI or ultrasound images with mammograms can increase detection sensitivity in dense breast tissue.

AI models trained on datasets primarily comprised of Caucasian individuals may be unable to generalize to Asian, African, or Hispanic populations. Differences in

Table 8 Datasets for breast cancer classification with weblinks

S.No.	Dataset name	Weblink(s)
1.	Wisconsin's	https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data
2.	SEER Dataset	https://ieee-dataport.org/open-access/seer-breast-cancer-data https://www.kaggle.com/datasets/ahmad03038/seer-breast-cancer-data-set
3.	Mammogram	https://www.cancerimagingarchive.net/collection/cbis-ddsm/ https://www.kaggle.com/datasets/awsaf49/cbis-ddsm-breast-cancer-image-dataset https://www.bcsc-research.org/index.php/datasets/mammography_dataset https://www.repository.cam.ac.uk/items/b6a97f0c-3b9b-40ad-8f18-3d121eef1459
4.	Ultrasound	https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset https://www.kaggle.com/datasets/sabahesaraki/breast-ultrasound-images-dataset https://qamebi.com/breast-ultrasound-images-database
5.	BreakHis	https://web.inf.ufpr.br/vri/databases/breast-cancer-histopathological-database-BreakHis/ https://data.mendeley.com/datasets/jxwvdwhpc2/1 https://www.kaggle.com/datasets/ambarish/BreakHis

mammography machines, configurations, and picture resolution among hospitals may also affect the efficacy of the AI models. Adding datasets from numerous locations and ethnic groups helps strengthen the model's resilience. AI programs may incorrectly categorize blurred images as benign or overlook subtle anomalies. Certain AI models exhibit optimal performance exclusively on specific machine types, resulting in subpar cross-device efficacy. Standardizing contrast, eliminating artifacts, and enhancing clarity through image pre-processing before building AI models may improve performance.

4.3 Reliance on Specific Datasets

Most studies utilize several prominent datasets, such as INbreast, DDSM, CBIS-DDSM, MIAS, or specific hospital data. These datasets exhibit insufficient geographic, ethnic, and demographic diversity, thus constraining the generalizability of the AI model. when an AI model is trained on a dataset that does not represent diverse populations may result in discrepancies in diagnostic precision, affects the model performance in accurately classifying the new or unseen instances, the reported performance metrics may not be reliable indicators, and limits the model's integration into wider clinical practice. Consequently, it would be advantageous to utilize datasets from several institutions and nations to encompass patients from various ethnic and genetic origins. Furthermore, promoting collaboration among hospitals and research institutes to exchange anonymized mammography, ultrasound, and MRI image data. Evaluate AI models using entirely separate datasets from numerous organizations and validate them in real-world hospital settings by contrasting AI predictions with radiologists' decisions.

4.4 Financial Implications of AI Integration in Breast Cancer Diagnosis

Incorporating Artificial Intelligence (AI) in breast cancer diagnostics offers economic advantages and challenges. Although AI can save expenses, increase efficiency, and augment diagnostic precision, the initial capital investment, ongoing maintenance, and regulatory obligations may be considerable.

- **Preliminary investment for AI deployment**.

To integrate AI models in clinical implementations, high-performance computing gear, software licenses, and cloud storage for managing extensive medical imaging data, together with high-quality AI models for massive, annotated datasets, will be necessary. Expert radiologists must manually annotate mammograms, ultrasounds, and MRIs to develop AI models, incurring significant costs. Clinical validation trials impose additional financial burdens.

- **Maintenance and enduring expenses**.

The AI model necessitates constant retraining with updated patient data to preserve accuracy, demanding continuous computational resources and human supervision. AI must interact easily with diverse hospital PACS, and modification for individual hospitals and locations incurs additional expenses. Neglecting to update AI models annually may lead to erroneous diagnoses and even legal repercussions.

- **Challenges faced by low-income nations**.

Insufficient funding and infrastructure hinder the deployment of AI in developing nations. Rural hospitals lack cloud storage and processing capacity to operate AI models effectively. Radiologists and technicians require training in artificial intelligence, hence increasing implementation

expenses. Nonetheless, the clinical use of AI alleviates the burden on radiologists by prioritizing high-risk instances. It can diminish false positives, hence decreasing superfluous biopsies and follow-up procedures. It also accelerates screening results in more patients diagnosed in a reduced timeframe, lowering operating expenses.

4.5 Regulatory and Compliance Considerations

In numerous jurisdictions, clinical AI systems are required to secure authorization, such as Food and Drug Administration (FDA) approval in the United States and Conformite Europeene (CE) marking in Europe. This procedure entails proving dependability, safety, and effectiveness. Health Insurance Portability and Accountability (HIPAA) in the United States or General Data Protection Regulation (GDPR) in Europe should safeguard patient data and be utilized for training or operating AI models. Clinical pictures and patient data must be sent and stored securely, employing stringent user authentication and maintaining audit trails. Continuous training or research data should be de-identified to safeguard patient privacy. Models trained on data lacking representation of varied populations may yield biased outcomes. However, it is crucial to assess performance among various demographic groups continuously.

5 Steps Involved in the BC Classification Utilizing Various Medical Imaging Modalities

5.1. *Data Collection and Preparation*.

- **Image Acquisition**: Acquire high-resolution images from various modalities (mammography, ultrasound, MRI, etc.). Guarantee uniform image quality and resolution across modalities.
- **Image Preprocessing**: Noise Reduction: Eliminate noise and artifacts from images with techniques such as filtering (e.g., Gaussian, median), wavelet denoising, or non-local means denoising.
- **Contrast Augmentation**: Enhance image contrast using histogram equalization, adaptive histogram equalization, or contrast-limited adaptive histogram equalization (CLAHE) to augment feature visibility.
- **Region of Interest (ROI) Extraction**: Isolate areas harbouring probable malignancies utilizing thresholding, region growth, active contours (snakes), or deep learning-based segmentation algorithms.
- **Image Normalization**: Standardize the intensity values of images across various modalities and patients to mitigate variances in image acquisition and enhance comparability.

5.2. *Feature Extraction*.
5.3. Derive meticulously designed features such as:

- **Geometric Attributes**: Area, perimeter, compactness, eccentricity, etc.
- **Texture Characteristics**: Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP), Gabor Filters, etc.
- **Radiomic Features**: First-order statistics (mean, standard deviation), higher-order statistics (skewness, kurtosis), wavelet-derived features, etc.
- **Automated Feature Extraction**: Employ deep learning architectures such as Convolutional Neural Networks (CNNs) to autonomously acquire distinguishing characteristics from images.

5.1 Feature Selection

Identify the most pertinent features that contribute to tumor classification. This can be accomplished by methods such as:

- **Filter Methods**: Correlation-based feature selection, information gain, chi-squared test.
- **Wrapper Methods**: Recursive Feature Elimination (RFE), forward feature selection, reverse feature selection.
- **Embedded Methods**: L1 regularization (LASSO), L2 regularization (Ridge).

5.2 Classification

Select an appropriate categorization algorithm, such as:

- **Machine Learning**: Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forests, Decision Trees.
- **Deep Learning**: Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and further deep architectures.
- **Training and Evaluation**: Train the selected classifier on a labeled dataset and assess its performance utilizing metrics including accuracy, sensitivity, specificity, precision, recall, F1-score, and AUC (Area Under the ROC Curve).

5.5. *Optimization of the Model*.

- **Hyperparameter Optimization**: Refine the hyperparameters of the selected classifier employing methods such as grid search, random search, or Bayesian optimization.
- **Ensemble Techniques**: Integrate various classifiers to enhance performance and resilience, including bagging, boosting, or stacking.
- **Data Augmentation**: Enhance the volume and variety of the training dataset by doing changes such as rotation, flipping, scaling, and cropping on the original photos.

5.3 Multi-modal Integration

Integrate data from many imaging modalities to enhance classification precision. This can be accomplished at the feature level, decision level, or through deep learning systems capable of integrating input from many modalities.

5.4 Deployment and Surveillance of the Model

- **Implement the Model**: Incorporate the trained model into a clinical decision support system.
- **Evaluate Performance**: Consistently assess the performance of the implemented model and regularly retrain it with updated data to ensure precision and adapt to evolving circumstances.

5.5 Ethical Considerations

- **Data Privacy and Security**: Safeguard the confidentiality and integrity of patient data during the entire procedure. Address any biases in the data and ensure the model is equitable and just across diverse patient populations.
- **Explainability and Interpretability**: Formulate techniques to elucidate the model's predictions to physicians, fostering trust and enhancing clinical decision-making.

5.6 Clinical Implementation of AI Models

The picture archiving and communications system (PACS) at the hospital constitutes an integral component of the Radiology department's infrastructure, facilitating communication among many elements, which include the radiological information system (RIS), electronic medical record (EMR), and imaging modalities. AI models may be incorporated as software modules or cloud-based services within the hospital network, automatically receiving mammograms or other breast images from the PACS, processing them, conducting training and validation, and transmitting findings (lesion heatmaps, suspicion scores, identified lesions) to the radiologist's workstation. This configuration replicates the dual-review procedure employed in numerous breast screening initiatives when a second radiologist typically re-evaluates the scans. Radiologists initially interpret the mammography and then verify the AI model's output to affirm or emphasize any missed abnormalities. Using AI (machine learning and deep learning) models in clinical processes toward breast cancer detection presents significant potential for enhancing efficiency and diagnostic precision. Thus, integrating AI outputs with the patient's electronic health record (EHR) may offer a more comprehensive assessment (e.g., risk factors, prior imaging, genetic predispositions). However, for radiologists to trust and use AI models in medical decision-making, they must be trained to effectively assess the AI outputs and comprehend whatever limits the model may have.

6 Future Research Directions

Here are promising future research directions for breast cancerous tumor classification using machine learning and deep learning algorithms.

6.1 Dataset Diversity and Integration

- Multi-Modal Data Fusion: Multi-modal data fusion combines imaging, including mammography, ultrasound, and MRI, with non-image sources like histopathological images to improve the accuracy and the reproducibility of the classification of breast cancerous tumor types.

Each image has its focus as follows:

- **Mammography**: Good for fibroglandular tissue and mass structures to observe microcalcifications.
- **Ultrasound** for distinguishing between cystic and solid lesions.
- **MRI** provides high-quality three-dimensional images and can visualize even minor tissue structure changes.
- **Histopathology**: best recognizes tumor structural characteristics morphologically at the cellular pattern.

Merging these types of complementary data allows deep learning to work with different feature sets, which is an advantage over using only one feature from single-modal approaches.

- **Inclusion of Genomic and Proteomic Data**: Using genomic and proteomic level data for domain-specific cancer prediction.
- **Cross-Dataset Generalization**: Teaching the models with several datasets (e.g., using publicly supported datasets MIAS, DDSM, CBIS-DDSM, BreakHis) to attain a wide range of coverage and less chances of the model being over-fitted.

6.2 Advanced Deep Learning Architectures

- **Hybrid Models**: CNNs, Transformers, and more conventional ML molding techniques for feature extraction and the classification phase of models.
  - **CNNs** excel at spatial feature extraction, especially for image and grid-like data.
  - **Transformers** perform well on tasks requiring sequence and attention, thus making them effective at capturing contextual relationships and long-range interdependence.
  - **Conventional machine learning algorithms**, such as Support Vector Machines (SVMs), Random Forests, and Gradient Boosting, are practical for tasks involving final classification or regression using various characteristics that have been retrieved.
- **Attention Mechanisms**: Targeting the most important areas in medical imaging using attention-based networks (spatial attention, channel attention, transformer-based self-attention, and hybrid attention).
  - **Subtle Features**: In breast cancer imaging, tumors and lesions may have faint or asymmetrical patterns.
  - **Region Focus**: Tumor locations are given priority by attention mechanisms, which disregard background noise.
  - **Interpretability**: Using attention maps, radiologists can validate results by highlighting the areas affecting the prediction.
  - **Efficiency**: Reduces computational overhead by focusing on critical areas.
- **Self-Supervised Learning**: Leveraging unlabelled data through pre-training and fine-tuning approaches.
  - **Contrastive Learning**: SimCLR and MoCo are examples of models that learn by differentiating between similar image pairings and those that are not.
  - **Masked Image Modeling**: Predict missing or masked parts of an image using MAE – Masked Autoencoders.
  - **Autoencoders**: Force models to capture latent feature representations by reconstructing input images.
  - **Rotation Prediction**: Predict the orientation of rotated images to gain an understanding of spatial context.
- **Fine-Tuning Phase (Labeled Data)**: adapting a pre-trained model to a downstream task with smaller labeled datasets is necessary.

6.1. *Model Explainability and Interpretability*.

- **Explainable AI (XAI)**: Enhancing transparency in model predictions with tools like Grad-CAM, LIME, and SHAP.
  - **Gradient-weighted Class Activation Mapping or Grad-CAM**: A bright red spot in a mammography image is highlighted by Grad-CAM, which indicates the tumor area that is responsible for a "Malignant" prediction.
  - **Local Interpretable Model-Agnostic Explanations, or LIME**, is a technique that helps identify critical areas in a breast MRI scan. Furthermore, we significantly impact a "Suspicious Lesion" prediction.
  - High pixel intensities in a particular area contribute the most to a "Malignant" prediction, according to SHAP (Shapley Additive exPlanations).
- **Reliable AI Models**: Make sure medical professionals comprehend and trust AI-driven diagnosis recommendations.

6.2. *Data Imbalance and Annotation Challenges*.

- **Mitigation of Class Imbalance**: Using loss functions (e.g., focal loss), synthetic data generation (e.g., GANs), and sophisticated resampling approaches.
- **Automated Annotation Tools**: AI-powered labelling enhances dataset quality and minimizes human mistakes.

6.3. *Privacy-Preserving and Federated Learning*.

- **Federated Learning (FL)**: Ensuring patient data privacy while training models across decentralized datasets.
- **Secure Multi-Party Computation (SMPC)**: Facilitating cooperative research without disclosing personal information.

6.4. *Risk Prediction and Early Detection*.

- **Analysing longitudinal data** allows for early intervention by following the development of tumors over time.
- **Predictive models**: creating instruments to calculate the likelihood of metastasis or recurrence.

6.5. *Tailored Therapy Programs*.

- **Precision Medicine Methods**: AI-based tailored treatment recommendations based on tumor classification.
- **Treatment Response Prediction**: ML/DL models predict how effective medicines will be.

6.6. *Real-World Implementation and Clinical Integration*.

- **Integration with Clinical Workflows**: Ensuring that ML/DL tools are adopted in diagnostic centers and hospitals.
- **Continuous Monitoring and Updates**: Models are periodically improved using fresh and actual data.

7 Conclusion

Breast cancer remains one of the leading causes of illness and death among women worldwide. Early and accurate detection is critical to improving outcomes and saving lives. This review focused on commonly used diagnostic techniques such as mammography, MRI, CT, PET scans, and histopathological imaging. While these methods are invaluable, they come with significant challenges, including radiation exposure, reliance on skilled radiologists, and difficulty in achieving accurate results, especially for women with dense breast tissue. Recent advancements in machine learning (ML) and deep learning (DL) technologies have shown tremendous potential in overcoming these limitations. These tools can analyze complex medical data and identify subtle patterns, enabling earlier and more precise detection of breast cancer, even under challenging imaging conditions. By integrating artificial intelligence with medical imaging, researchers have made significant progress in detecting cancer at its earliest stages.

Our review explored the performance of various ML and DL models applied to key datasets like Wisconsin, SEER, mammography images, ultrasound scans, and BreakHis histopathology images. Models such as convolutional neural networks (CNNs), ensemble methods, and hybrid approaches have achieved impressive results, with some reaching accuracy levels above 99%. These advancements have improved diagnostic accuracy, reduced reliance on human interpretation, and helped distinguish between benign and malignant tumors more effectively. However, challenges remain. Issues such as imbalanced datasets, noisy imaging data, and limited diversity in patient data raise concerns about the reliability and generalizability of AI models. Additionally, creating robust models requires vast amounts of labeled data, particularly for histological and radiological image classification. Privacy concerns and the lack of external validations further complicate the adoption of AI technologies in clinical settings.

New approaches such as federated learning, better feature selection techniques, and synthetic data generation are being explored to address these challenges. These methods aim to improve model performance while ensuring ethical and secure handling of sensitive patient data. Tools like Grad-CAM and LIME also play a crucial role by making AI decisions more transparent and understandable for clinicians, fostering trust in these technologies. These efforts demonstrate that AI-based solutions can complement traditional diagnostic methods, offering faster and more accurate cancer detection with minimal risks from invasive procedures or radiation exposure. However, moving from research to real-world application requires lightweight models for use in resource-limited settings, along with thorough external validation and real-world testing.

Author Contributions Vishnu Vardhana Reddy Karna: writing–review & editing, writing–original draft, conceptualization, investigation, prepared tables. Viswavardhan Reddy Karna: writing–original draft, writing–review & editing, prepared figures. Ravinder Beemagani: writing–review & editing, prepared tables. Varaprasad Janamala: writing –review & editing, prepared tables. V N Koteswara Rao Devana: writing–review & editing, formal analysis. S Rajasekhar K: formal analysis, prepared tables. V Ravi Sankar Ch: writing–review & editing, formal analysis, prepared figures. Vijayakumar Padarti: formal analysis, prepared tables.

Funding This work received no external funding.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

1. Visser LL, Groen EJ, van Leeuwen FE, Lips EH, Schmidt MK, Wesseling J (2019) Predictors of an invasive breast cancer recurrence after DCIS: a systematic review and meta-analyses. Cancer Epidemiol Biomarkers Prev 28(5):835–845. https://doi.org/10.1158/1055-9965.EPI-18-0976
2. ud din NM, Dar RA, Rasool M, Assad A (2022) Breast cancer detection using deep learning: Datasets, methods, and challenges

ahead, Oct. 01, *Elsevier Ltd*. https://doi.org/10.1016/j.compbiomed.2022.106073
3. Botlagunta M et al (2023) Classification and diagnostic prediction of breast cancer metastasis on clinical data using machine learning algorithms. Sci Rep. https://doi.org/10.1038/s41598-023-27548-w
4. Wu J, Hicks C (2021) Breast cancer type classification using machine learning. J Pers Med 11(2):1–12. https://doi.org/10.3390/jpm11020061
5. Karna VVR, Karna VR, Janamala V, Devana VNKR, Ch VRS, Tummala AB (2024) A comprehensive review on heart disease risk prediction using machine learning and deep learning algorithms. Springer Sci Bus Media B V. https://doi.org/10.1007/s11831-024-10194-4
6. Rama Sree S, Vanathi A, Veluri RK, Ramesh SNSVSC (2022) A comparative study on a disease prediction system using machine learning algorithms. https://doi.org/10.1007/978-3-030-95711-7_41
7. Thirumalaisamy S et al (2023) Breast cancer classification using synthesized deep learning model with metaheuristic optimization algorithm. Diagnostics. https://doi.org/10.3390/diagnostics13182925
8. Khalid A et al (2023) Breast cancer detection and prevention using machine learning. Diagnostics. https://doi.org/10.3390/diagnostics13193113
9. Anuradha Bharadvaja N (2023) Exploring different computational approaches for effective diagnosis of breast cancer. Prog Biophys Mol Biol 177:141–150. https://doi.org/10.1016/j.pbiomolbio.2022.11.004
10. Thorneloe RJ et al (2019) Beliefs about medication and uptake of preventive therapy in women at increased risk of breast cancer: results from a multicenter prospective study. Clin Breast Cancer 19(1):e116–e126. https://doi.org/10.1016/j.clbc.2018.10.008
11. Breast cancer, https://www.who.int/news-room/fact-sheets/detail/breast-cancer
12. Breast Cancer Facts and Statistics https://www.breastcancer.org/facts-statistics
13. Rautela K, Kumar D, Kumar V (2022) A systematic review on breast cancer detection using deep learning techniques. Arch Computat Methods Eng. https://doi.org/10.1007/s11831-022-09744-5
14. Paszat L, Sutradhar R, Zhou L, Nofech-Mozes S, Rakovitch E (2019) Including the ductal Carcinoma-In-Situ (DCIS) score in the development of a multivariable prediction model for recurrence after excision of DCIS. Clin Breast Cancer. 19(1): 35–46. https://doi.org/10.1016/j.clbc.2018.07.018
15. Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A (2018) Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 68(6):394–424. https://doi.org/10.3322/caac.21492
16. Liu L, Hao X, Song Z, Zhi X, Zhang S, Zhang J (2021) Correlation between family history and characteristics of breast cancer. Sci Rep. https://doi.org/10.1038/s41598-021-85899-8
17. Sathishkumar K, Chaturvedi M, Das P, Stephen S, Mathur P (2022) Cancer incidence estimates for 2022 & projection for 2025: result from National Cancer Registry Programme, India. Indian J Med Res 156(4):598–607. https://doi.org/10.4103/ijmr.ijmr_1821_22
18. Orrantia-Borunda E, Anchondo-Nuñez P, Acuña-Aguilar LE, Gómez-Valles FO, Ramírez-Valdespino CA (2022) Subtypes of Breast cancer. in Breast cancer. Exon, pp 31–42. doi: doi: https://doi.o rg/10.36255/exon-publications-breast-cancer-subtypes
19. Łukasiewicz S, Czeczelewski M, Forma A, Baj J, Sitarz R, Stanisławek A (2021) Breast cancer—epidemiology, risk factors, classification, prognostic markers, and current treatment strategies—an updated review. Cancers. https://doi.org/10.3390/cancers13174287
20. Makki J (2015) Diversity of breast carcinoma: histological subtypes and clinical relevance. Clin Med Insights Pathol. https://doi.org/10.4137/CPath.S31563
21. Raj-Kumar P-K et al (2019) PCA-PAM50 improves consistency between breast cancer intrinsic and clinical subtyping reclassifying a subset of luminal A tumors as luminal B. Sci Rep 9(1): 7956. https://doi.org/10.1038/s41598-019-44339-4
22. Wang J, Xu B (2019) Targeted therapeutic options and future perspectives for HER2-positive breast cancer. Signal Transduct Target Ther 4(1):34. https://doi.org/10.1038/s41392-019-0069-2
23. Aznab M, Shojae S, Golpazir Sorkheh A, Eslam pia K, Rezaei M (2022) The survival of patients with triple negative breast cancer undergoing chemotherapy along with lifestyle change interventions. Arch Breast Cancer 10(1):66–73. https://doi.org/10.32768/abc.202310166-73
24. Łukasiewicz S, Czeczelewski M, Forma A, Baj J, Sitarz R, Stanisławek A (2021) Breast Cancer—Epidemiology, risk factors, classification, prognostic markers, and current treatment Strategies—An updated review. Cancers (Basel) 13(17): 4287. https://doi.org/10.3390/cancers13174287
25. Giaquinto AN et al (2024) Breast cancer statistics 2024. CA Cancer J Clin. https://doi.org/10.3322/caac.21863
26. NBCF team Breast Cancer Facts & Stats. https://www.nationalbreastcancer.org/breast-cancer-facts/
27. Zhu JW, Charkhchi P, Adekunte S, Akbari MR (1917) What is known about breast cancer in young women? *Cancers (Basel)*, https://doi.org/10.3390/cancers15061917
28. Sun YS et al (2017) Risk factors and preventions of breast cancer. Int J Biol Sci. https://doi.org/10.7150/ijbs.21635
29. Momenimovahed Z, Salehiniya H (2019) Epidemiological characteristics of and risk factors for breast cancer in the world. Dove Med Press Ltd. https://doi.org/10.2147/BCTT.S176070
30. Chen D, Huang M, Li W (2021) Knowledge-powered deep breast tumor classification with multiple medical reports. IEEE ACM Trans Comput Biol Bioinform 18(3):891–901. https://doi.org/10.1109/TCBB.2019.2955484
31. Xu J et al (2016) Stacked sparse autoencoder (SSAE) for nuclei detection on breast cancer histopathology images. IEEE Trans Med Imaging 35(1): 119–130. https://doi.org/10.1109/TMI.2015.2458702
32. Trayes KP, Cokenakes SEH Breast Cancer Treatment., *Am Fam Physician*, vol. 104, no. 2, pp. 171–178, Aug. 2021, [Online]. Available: http://www.ncbi.nlm.nih.gov/pubmed/34383430
33. Barazi H, Gunduru M (2024) *Mammography BI RADS Grading*. StatPearls
34. Fico N et al (2023) Breast Imaging Physics in Mammography (Part I), Oct. 01, *Multidisciplinary Digital Publishing Institute (MDPI)*. https://doi.org/10.3390/diagnostics13203227
35. Radhakrishna S et al (2018) Role of magnetic resonance imaging in breast cancer management. South Asian J Cancer 07(02):069–071. https://doi.org/10.4103/sajc.sajc_104_18
36. Chang R-F, Chen H-H, Chang Y-C, Huang C-S, Chen J-H, Lo C-M (2016) Quantification of breast tumor heterogeneity for ER status, HER2 status, and TN molecular subtype evaluation on DCE-MRI. Magn Reson Imaging 34(6):809–819. https://doi.org/10.1016/j.mri.2016.03.001
37. Lee J, Kim SH, Kang BJ, Lee A, Park WC, Hwang J (2021) Imaging characteristics of young age breast cancer (YABC) focusing on pathologic correlation and disease recurrence. Sci Rep. https://doi.org/10.1038/s41598-021-99600-6
38. Hill DA et al (2018) Utilization of breast cancer screening with magnetic resonance imaging in community practice. J Gen Intern Med 33(3):275–283. https://doi.org/10.1007/s11606-017-4224-6

39. Amioka A et al (2016) Ability of contrast-enhanced ultrasonography to determine clinical responses of breast cancer to neoadjuvant chemotherapy. Jpn J Clin Oncol 46(4): 303–309. https://doi.org/10.1093/jjco/hyv215
40. Nicosia L et al (2020) Automatic breast ultrasound: state of the Art and future perspectives. Ecancermedicalscience. https://doi.org/10.3332/ECANCER.2020.1062
41. Groheux D, Cochet A, Humbert O, Alberini J-L, Hindié E, Mankoff D (2016) F-FDG PET/CT for staging and restaging of breast cancer. J Nucl Med 57(Supplement 1):17S-26S. https://doi.org/10.2967/jnumed.115.157859
42. Cazzato RL et al (2018) PET/CT-guided interventions: indications, advantages, disadvantages and the state of the Art. Minim Invasive Therapy Allied Technol. 27(1): 27–32. https://doi.org/10.1080/13645706.2017.1399280
43. Park J, Kim C, Cha YK, Chung MJ (May2023) Missed and detected incidental breast cancers on contrast enhanced chest CT: detection rates and CT features. Diagnostics. https://doi.org/10.3390/diagnostics13091522
44. Gilbert FJ, Pinker-Domenig K (2019) Diagnosis and staging of breast cancer: when and how to use mammography, tomosynthesis, ultrasound, Contrast-Enhanced mammography, and magnetic resonance imaging. https://doi.org/10.1007/978-3-030-11149-6_13
45. Baltzer PAT, Kapetas P, Marino MA, Clauser P (2017) New diagnostic tools for breast cancer. memo - Magazine Eur Med Oncol 10(3):175–180. https://doi.org/10.1007/s12254-017-0341-5
46. Lowry KP et al (2020) Long-term outcomes and cost-effectiveness of breast cancer screening with digital breast tomosynthesis in the United States. JNCI J Natl Cancer Inst 112(6):582–589. https://doi.org/10.1093/jnci/djz184
47. Mumin NA et al (2019) Diagnostic efficacy of synthesized 2D digital breast tomosynthesis in multi-ethnic Malaysian population. Sci Rep. https://doi.org/10.1038/s41598-018-37451-4
48. Zapater-Moros A et al (2018) Probabilistic graphical models relate immune status with response to neoadjuvant chemotherapy in breast cancer. Oncotarget 9(45):27586–27594. https://doi.org/10.18632/oncotarget.25496
49. Zuo D, Yang L, Jin Y, Qi H, Liu Y, Ren L (2023) Machine learning-based models for the prediction of breast cancer recurrence risk. BMC Med Inf Decis Mak 23(1): 276. https://doi.org/10.1186/s12911-023-02377-z
50. Jiang F et al (2017) Artificial intelligence in healthcare: past, present and future. Stroke Vasc Neurol 2(4):230–243. https://doi.org/10.1136/svn-2017-000101
51. Daimiel Naranjo I et al (2021) Radiomics and machine learning with multiparametric breast MRI for improved diagnostic accuracy in breast cancer diagnosis. Diagnostics 11(6):919. https://doi.org/10.3390/diagnostics11060919
52. Huang S, Yang J, Fong S, Zhao Q (2020) Artificial intelligence in cancer diagnosis and prognosis: opportunities and challenges. Cancer Lett 471: 61–71. https://doi.org/10.1016/j.canlet.2019.12.007
53. Wolberg W, Mangasarian O, Street N, Street W (1993) Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5DW2B
54. Naik AK, Kuppili V, Edla DR (2020) Efficient feature selection using one-pass generalized classifier neural network and binary bat algorithm with a novel fitness function. Soft Comput 24(6):4575–4587. https://doi.org/10.1007/s00500-019-04218-6
55. Lahoura V et al (2021) Cloud computing-based framework for breast cancer diagnosis using extreme learning machine. Diagnostics 11(2):241. https://doi.org/10.3390/diagnostics11020241
56. Alshayeji MH, Ellethy H, Abed S, Gupta R (2022) Computer-aided detection of breast cancer on the Wisconsin dataset: an artificial neural networks approach. Biomed Signal Process Control. https://doi.org/10.1016/j.bspc.2021.103141
57. Kadhim RR, Kamil MY (2023) Comparison of machine learning models for breast cancer diagnosis. IAES Int J Artif Intell 12(1):415–421. https://doi.org/10.11591/ijai.v12.i1.pp415-421
58. Singh D, Nigam R, Mittal R, Nunia M (2023) Information retrieval using machine learning from breast cancer diagnosis. Multimed Tools Appl. 82(6): 8581–8602. https://doi.org/10.1007/s11042-022-13550-3
59. Elsadig MA, Altigani A, Elshoush HT (2023) Breast cancer detection using machine learning approaches: a comparative study. Int J Electr Comput Eng. 13(1): 736–745. https://doi.org/10.11591/ijece.v13i1.pp736-745
60. Al Reshan MS et al (2023) Enhancing breast cancer detection and classification using advanced Multi-Model features and ensemble machine learning techniques. Life. https://doi.org/10.3390/life13102093
61. Mushtaq Z, Qureshi MF, Abbass MJ, Al-Fakih SMQ (Jan.2023) Effective kernel-principal component analysis based approach for Wisconsin breast cancer diagnosis. Electron Lett. https://doi.org/10.1049/ell2.12706
62. Kaur H (2023) Dense convolutional neural network based deep learning framework for the diagnosis of breast cancer. Wirel Pers Commun 132(3):1765–1780. https://doi.org/10.1007/s11277-023-10678-9
63. Singh LK, Khanna M, Singh R (2024) An enhanced soft-computing based strategy for efficient feature selection for timely breast cancer prediction: Wisconsin Diagnostic Breast Cancer dataset case. Multimed Tools Appl 83(31):76607–76672. https://doi.org/10.1007/s11042-024-18473-9
64. Awan MZ, Arif MS, Abideen MZU, Abodayeh K (2024) Comparative analysis of machine learning models for breast cancer prediction and diagnosis: A dual-dataset approach. Indonesian J Electr Eng Comput Sci. 34(3): 2032–2044. https://doi.org/10.11591/ijeecs.v34.i3.pp2032-2044
65. Jony AI, Arnob AKB (2024) Deep learning paradigms for breast cancer diagnosis: a comparative study on Wisconsin diagnostic dataset. Malaysian J Sci Adv Technol. https://doi.org/10.56532/mjsat.v4i2.245
66. Jing, Teng SEER Breast Cancer Data, https://ieee-dataport.org/open-access/seer-breast-cancer-data. https://doi.org/10.21227/a9qy-ph35
67. Pan X, Yang W, Chen Y, Tong L, Li C, Li H (2019) Nomogram for predicting the overall survival of patients with inflammatory breast cancer: a SEER-based study. Breast 47:56–61. https://doi.org/10.1016/j.breast.2019.05.015
68. Shukla N, Hagenbuchner M, Win KT, Yang J (2018) Breast cancer data analysis for survivability studies and prediction. Comput Methods Programs Biomed. 155:199–208. https://doi.org/10.1016/j.cmpb.2017.12.011
69. Fotouhi S, Asadi S, Kattan MW (2019) A comprehensive data level analysis for cancer diagnosis on imbalanced data. J Biomed Inf 90. https://doi.org/10.1016/j.jbi.2018.12.003
70. Abdikenov B, Iklassov Z, Sharipov A, Hussain S, Jamwal PK (2019) Analytics of heterogeneous breast cancer data using neuroevolution. IEEE Access 7:18050–18060. https://doi.org/10.1109/ACCESS.2019.2897078
71. Simsek S, Kursuncu U, Kibis E, AnisAbdellatif M, Dag A (2020) A hybrid data mining approach for identifying the temporal effects of variables associated with breast cancer survival. Expert Syst Appl. https://doi.org/10.1016/j.eswa.2019.112863
72. Salehi M, Razmara J, Lotfi S (2020) A novel data mining on breast cancer survivability using MLP ensemble learners. Comput J 63(3):435–447. https://doi.org/10.1093/comjnl/bxz051
73. Gupta S, Gupta MK (2022) A comparative analysis of deep learning approaches for predicting breast cancer survivability. Arch

Comput Methods Eng 29(5):2959–2975. https://doi.org/10.1007/s11831-021-09679-3
74. Li C et al (2023) Novel models by machine learning to predict prognosis of breast cancer brain metastases. J Transl Med. https://doi.org/10.1186/s12967-023-04277-2
75. Wu R et al (2023) Evaluation of machine learning algorithms for the prognosis of breast cancer from the Surveillance, Epidemiology, and End Results database. PLoS One. https://doi.org/10.1371/journal.pone.0280340
76. Manikandan P, Durga U, Ponnuraja C (2023) An integrative machine learning framework for classifying SEER breast cancer. Sci Rep. https://doi.org/10.1038/s41598-023-32029-1
77. Kolasseri AE (2024) Comparative study of machine learning and statistical survival models for enhancing cervical cancer prognosis and risk factor assessment using SEER data. Sci Rep 14(1):22203. https://doi.org/10.1038/s41598-024-72790-5
78. Esmaeili M, Ayyoubzadeh SM, Ahmadinejad N, Ghazisaeedi M, Nahvijou A, Maghooli K (2020) A decision support system for mammography reports interpretation. Health Inf Sci Syst 8(1):17. https://doi.org/10.1007/s13755-020-00109-5
79. Chotai N, Kulkarni S (2020) General considerations in breast imaging. in Breast imaging essentials. Springer Singapore, Singapore, pp 3–8. doi: https://doi.org/10.1007/978-981-15-1412-8_1.
80. Abbasian Ardakani A, Mohammadi A, Mirza-Aghazadeh-Attari M, Acharya UR (2023) An open-access breast lesion ultrasound image database: applicable in artificial intelligence studies. Comput Biol Med. https://doi.org/10.1016/j.compbiomed.2022.106438
81. Abdel-Nasser M, Melendez J, Moreno A, Omer OA, Puig D (2017) Breast tumor classification in ultrasound images using texture analysis and super-resolution methods. Eng Appl Artif Intell 59:84–92. https://doi.org/10.1016/j.engappai.2016.12.019
82. Qi X et al (2019) Automated diagnosis of breast ultrasonography images using deep neural networks. Med Image Anal 52:185–198. https://doi.org/10.1016/j.media.2018.12.006
83. Moon WK et al (2020) Computer-aided tumor detection in automated breast ultrasound using a 3-D convolutional neural network. Comput Methods Programs Biomed. https://doi.org/10.1016/j.cmpb.2020.105360
84. Pavithra S, Vanithamani R, Justin J (2020) Computer-aided breast cancer detection using ultrasound images. Mater Today Proc. https://doi.org/10.1016/j.matpr.2020.08.381
85. Wang Y, Choi EJ, Choi Y, Zhang H, Jin GY, Ko SB (May2020) Breast cancer classification in automated breast ultrasound using multiview convolutional neural network with transfer learning. Ultrasound Med Biol 46(5):1119–1132. https://doi.org/10.1016/j.ultrasmedbio.2020.01.001
86. Shia WC, Chen DR (2021) Classification of malignant tumors in breast ultrasound using a pre-trained deep residual network model and support vector machine. Comput Med Imaging Graph. https://doi.org/10.1016/j.compmedimag.2020.101829
87. Shia WC, Lin LS, Chen DR (2021) Classification of malignant tumors in breast ultrasound using unsupervised machine learning approaches. Sci Rep. https://doi.org/10.1038/s41598-021-81008-x
88. Ayana G, Park J, Jeong JW, Choe SW (2022) A novel multistage transfer learning for ultrasound breast cancer image classification. Diagnostics 12(1):135. https://doi.org/10.3390/diagnostics12010135
89. Chen S-H, Wu Y-L, Pan C-Y, Lian L-Y, Su Q-C (Sep.2023) Breast ultrasound image classification and physiological assessment based on GoogLeNet. J Radiat Res Appl Sci. 16(3). https://doi.org/10.1016/j.jrras.2023.100628
90. Alotaibi M et al (2023) Breast cancer classification based on convolutional neural network and image fusion approach using ultrasound images, *Heliyon.* https://doi.org/10.1016/j.heliyon.2023.e22406
91. Abdel-Nasser M, Rashwan HA, Puig D, Moreno A (2015) Analysis of tissue abnormality and breast density in mammographic images using a uniform local directional pattern. Expert Syst Appl 42(24):9499–9511. https://doi.org/10.1016/j.eswa.2015.07.072
92. Sadad T, Munir A, Saba T, Hussain A (2018) Fuzzy c-means and region growing based classification of tumor from mammograms using hybrid texture feature. J Comput Sci 29:34–45. https://doi.org/10.1016/j.jocs.2018.09.015
93. Shen L, Margolies LR, Rothstein JH, Fluder E, McBride R, Sieh W (2019) Deep learning to improve breast cancer detection on screening mammography. Sci Rep. https://doi.org/10.1038/s41598-019-48995-4
94. Vijayarajeswari R, Parthasarathy P, Vivekanandan S, Basha AA (2019) Classification of mammogram for early detection of breast cancer using SVM classifier and Hough transform. Meas Lond 146:800–805. https://doi.org/10.1016/j.measurement.2019.05.083
95. Huang ML, Lin TY (2021) Considering breast density for the classification of benign and malignant mammograms. Biomed Signal Process Control. https://doi.org/10.1016/j.bspc.2021.102564
96. Muduli D, Dash R, Majhi B (2021) Fast discrete curvelet transform and modified PSO based improved evolutionary extreme learning machine for breast cancer detection. Biomed Signal Process Control. https://doi.org/10.1016/j.bspc.2021.102919
97. Maqsood S, Damaševičius R, Maskeliūnas R (2022) TTCNN: a breast cancer detection and classification towards computer-aided diagnosis using digital mammography in early stages. Appl Sci. https://doi.org/10.3390/app12073273
98. Pattanaik RK, Mishra S, Siddique M, Gopikrishna T, Satapathy S (2022) Breast cancer classification from mammogram images using extreme learning machine-based DenseNet121 model. J Sens. https://doi.org/10.1155/2022/2731364
99. Jabeen K et al (2023) BC2NetRF: breast cancer classification from mammogram images using enhanced deep learning features and equilibrium-jaya controlled regula falsi-based features selection. Diagnostics. https://doi.org/10.3390/diagnostics13071238
100. Melekoodappattu JG, Dhas AS, Kandathil BK, Adarsh KS (2023) Breast cancer detection in mammogram: combining modified CNN and texture feature-based approach. J Ambient Intell Humaniz Comput 14(9):11397–11406. https://doi.org/10.1007/s12652-022-03713-3
101. Chugh G, Kumar S, Singh N (2024) TransNet: a comparative study on breast carcinoma diagnosis with classical machine learning and transfer learning paradigm. Multimed Tools Appl. 83(11):33855–33877. https://doi.org/10.1007/s11042-023-16938-x
102. Liu Z, Peng J, Guo X, Chen S, Liu L (Jun.2024) Breast cancer classification method based on improved VGG16 using mammography images. J Radiat Res Appl Sci 17(2):100885. https://doi.org/10.1016/j.jrras.2024.100885
103. Spanhol FA, Oliveira LS, Petitjean C, Heutte L Breast cancer histopathological image classification using Convolutional Neural Networks, in (2016) *International Joint Conference on Neural Networks (IJCNN)*, IEEE, Jul. 2016, pp. 2560–2567. https://doi.org/10.1109/IJCNN.2016.7727519
104. Spanhol FA, Oliveira LS, Cavalin PR, Petitjean C, Heutte L (2017) Deep features for breast cancer histopathological image classification, 2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Banff, AB, Canada, 2017, pp 1868-1873. https://doi.org/10.1109/SMC.2017.8122889
105. Al Nahid A, Mehrabi MA, Kong Y (2018) Histopathological breast cancer image classification by deep neural network

techniques guided by local clustering. Biomed Res Int. https://doi.org/10.1155/2018/2362108
106. Kumar K, Rao ACS Breast cancer classification of image using convolutional neural network, in (2018) *4th International Conference on Recent Advances in Information Technology (RAIT)*, IEEE, Mar. 2018, pp. 1–6. https://doi.org/10.1109/RAIT.2018.8389034
107. Deniz E, Şengür A, Kadiroğlu Z, Guo Y, Bajaj V, Budak Ü (2018) Transfer learning based histopathologic image classification for breast cancer detection. Health Inf Sci Syst. https://doi.org/10.1007/s13755-018-0057-x
108. Gour M, Jain S, Sunil Kumar T (2020) Residual learning based CNN for breast cancer histopathological image classification. Int J Imaging Syst Technol. 30(3):621–635. https://doi.org/10.1002/ima.22403
109. Yamlome P, Akwaboah AD, Marz A, Deo M, Convolutional Neural Network Based Breast Cancer Histopathology Image Classification, in (2020) *42nd Annual International Conference of the IEEE Engineering in Medicine & Biology* Society *(EMBC)*, IEEE, Jul. 2020, pp. 1144–1147. https://doi.org/10.1109/EMBC44109.2020.9176594
110. Uppada R, Pedada S, Chinni H Novel Neural Network for Breast Cancer Diagnosis, in (2023) *5th International Conference on Electrical, Computer and Communication Technologies, ICECCT 2023*, Institute of Electrical and Electronics Engineers Inc., 2023. https://doi.org/10.1109/ICECCT56650.2023.10179826
111. Wang W, Li Y, Yan X, Xiao M, Gao M (2024) Breast Cancer Image Classification Method Based on Deep Transfer Learning, Apr. [Online]. Available: http://arxiv.org/abs/2404.09226
112. Guo J et al (2024) A novel breast cancer image classification model based on multiscale texture feature analysis and dynamic learning. Sci Rep. https://doi.org/10.1038/s41598-024-57891-5
113. Balasubramanian AA et al (2024) Ensemble deep learning-based image classification for breast cancer subtype and invasiveness diagnosis from whole slide image histopathology. Cancers (Basel). https://doi.org/10.3390/cancers16122222

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.